

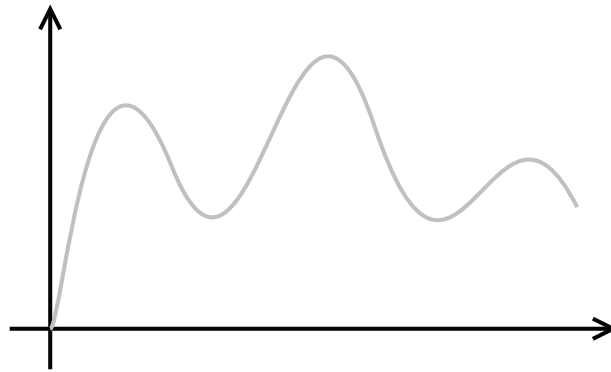
ECE 5200 Syllabus Highlights:

- Course materials at www.carmen.osu
 - syllabus
 - lecture packets
 - * authored by P. Schniter, L. Potter, K. Lee
 - supplementary notes
 - scanned lecture notes
 - assignments and solutions
 - sample test questions
 - links to online resources
- Office hours: TBA (Please respond to a scheduling poll.)
- Text: There exists an equivalence class of good texts
- Classroom goals: provide perspective, context, synthesis, emphases, and structure to make learning efficient
- Email: lee.8763@osu.edu

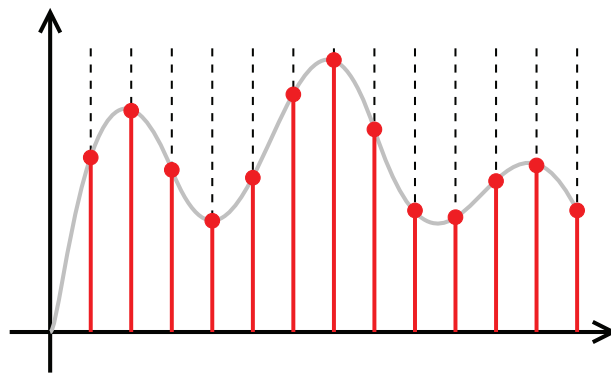
Introduction:

- Much of modern engineering is concerned with signals and systems.
- Roughly, a *signal* is an information-containing waveform (e.g., audio signal, image, digital video stream) and a *system* converts one waveform to another (e.g., low pass filter, mp3 encoder, a simple delay).
- Many real-world signals live in a *continuous* domain (e.g., continuous-time), while much of modern signal processing is done via computation in a *discrete* domain.
- In this course, we develop a solid understanding of signals/systems, especially the relationship between their continuous and discrete representations. The frequency domain will be of particular importance here.
- This course is about understanding/applying *concepts*, not memorizing formulas, and *mathematics* is the language of these concepts. Thus, a major goal of this course is making you comfortable with the mathematics of signals and systems.

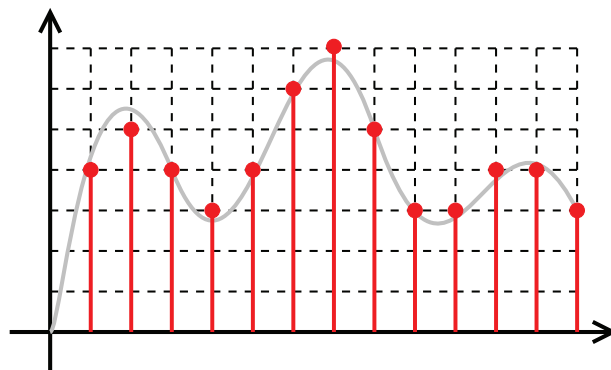
Continuous, Discrete-time, and Digital:



continuous-time signal, $x(t)$



sampled signal (discrete-time), $x[n] = x(nT)$

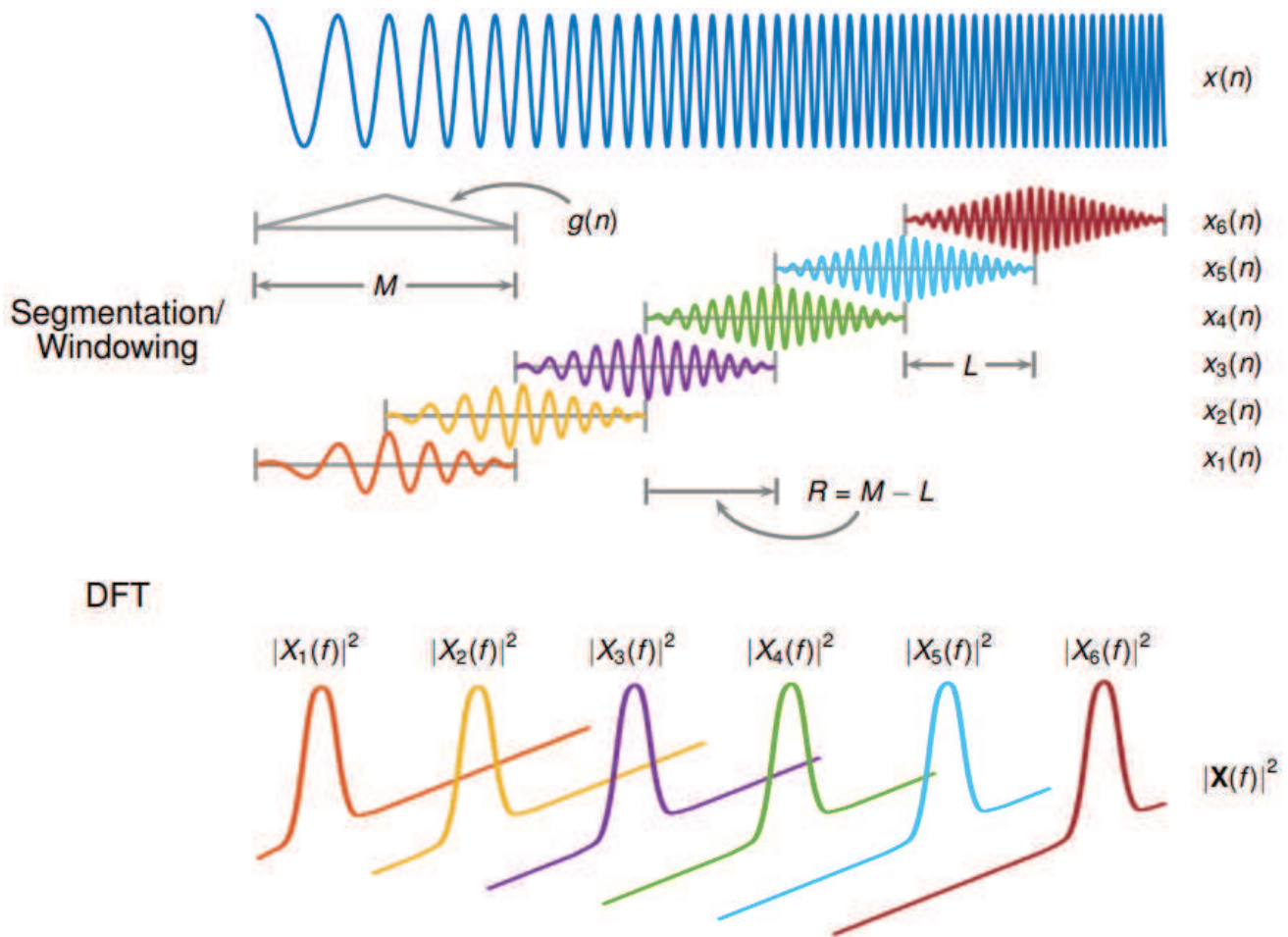


digital signal (discrete-time & quantized)

Outline:

1. Review of Signals/Systems Background
 - (a) Systems: linear, time-invariant, causal, stable
 - (b) Continuous-time transforms: FS, CTFT, Laplace
 - (c) Discrete-time transforms: DTFT, Z-transform
2. Sampling and Reconstruction
 - (a) sampling, aliasing, reconstruction (sinc & practical)
 - (b) upsampling, downsampling, and rate conversion
3. Processing of Finite-Length Signals
 - (a) DFT, circ conv, windowing, spectral analysis
 - (b) matrix/vector formulations
 - (c) FFT, fast convolution, overlap/save
4. Design of Discrete-Time Filters
 - (a) goals: magnitude, group delay, joint
 - (b) FIR designs: window, minimax, least-squares
 - (c) IIR designs: bilinear transform, Prony's method
5. Short-Time Fourier Transform, Filterbanks, Wavelets
6. Sparse Signal Processing and Inverse Problems

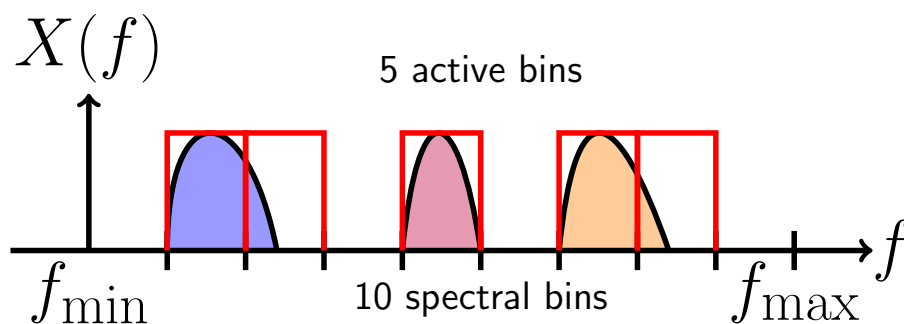
Time-Frequency Representations of Signals:



short-time Fourier transform of a chirp signal

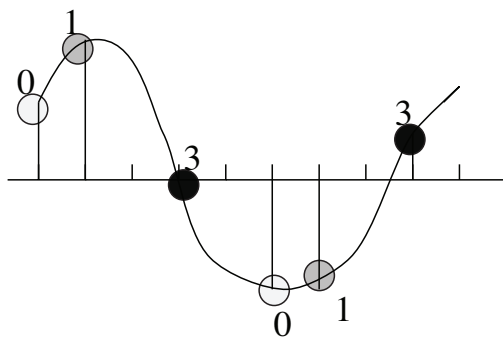
<https://www.mathworks.com/help/signal/ref/stft.html>

Nonuniform Sampling of Multiband Signals:



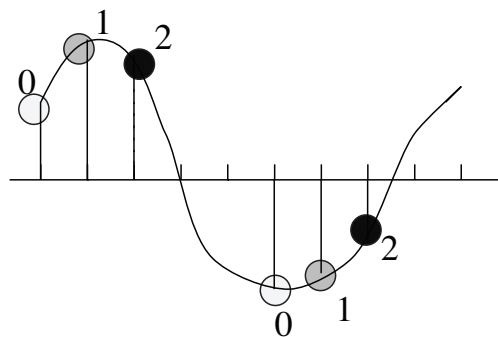
Multicoset sampling

Cosets = 0, 1, 3



Bunched sampling

Cosets = 0, 1, 2



Comments On Homework:

- **Learn by doing.** Learning and retention require repeated access and use of new ideas. Most learning in this course will come from reviewing notes and working through homework exercises. See links at Carmen for study tips.
- Suggested approach:
 1. Review lecture notes. And, as needed, consult textbooks, online resources, and Q&A sessions. Seek multiple explanations and viewpoints.
 2. Attempt to review and understand material *before* attempting the homework exercises.
 3. Make use of Q&A sessions and email for questions.
- *Caution:* Don't short-cut the homework! If you google for answers, or dig through homeworks from previous terms,
 1. You won't learn the material.
 2. You won't do well on the exams.

Complex Numbers:

- What are they?
 - Basically, a complex number is a pair of real numbers
 - Essential to many transforms (Fourier, Laplace, Z)
- Rectangular and Polar notation (say $x \in \mathbb{C}$):

$$x = x_r + jx_i$$

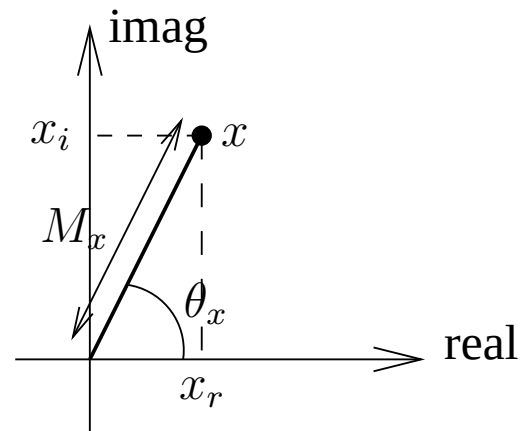
$$\text{for } x_r, x_i \in \mathbb{R}, \quad j \triangleq \sqrt{-1}$$

$$= M_x e^{j\theta_x}$$

$$\text{for } M_x \geq 0, \quad \theta_x \in [0, 2\pi)$$

$$= M_x (\cos \theta_x + j \sin \theta_x) \quad \text{via Euler's decomposition}$$

$$= \sqrt{x_r^2 + x_i^2} e^{j \arctan \frac{x_i}{x_r}}$$



- Arithmetic: (say $x, y \in \mathbb{C}$)
 - Sum:
$$x + y = (x_r + jx_i) + (y_r + jy_i) = (x_r + y_r) + j(x_i + y_i)$$
 - Product:
$$xy = (M_x e^{j\theta_x})(M_y e^{j\theta_y}) = M_x M_y e^{j(\theta_x + \theta_y)}$$
 - Complex conjugate:
$$x^* = x_r - jx_i = M_x e^{-j\theta_x}$$

Euler's Identity:

$$\begin{aligned}e^{j\theta} &= \cos \theta + j \sin \theta \\ \Rightarrow \cos \theta &= \frac{e^{j\theta} + e^{-j\theta}}{2} \\ \sin \theta &= \frac{e^{j\theta} - e^{-j\theta}}{2j}\end{aligned}$$

Why? Use series expansions (Madhava c. 1370; Taylor 1715; Maclaurin c. 1742)

$$\begin{aligned}e^z &= 1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \frac{z^4}{4!} + \frac{z^5}{5!} + \frac{z^6}{6!} + \dots \\ e^{j\theta} &= 1 + j\theta + \frac{(j\theta)^2}{2!} + \frac{(j\theta)^3}{3!} + \frac{(j\theta)^4}{4!} + \frac{(j\theta)^5}{5!} + \frac{(j\theta)^6}{6!} + \dots \\ &= 1 + j\theta - \frac{\theta^2}{2!} - j\frac{\theta^3}{3!} + \frac{\theta^4}{4!} + j\frac{\theta^5}{5!} - \frac{\theta^6}{6!} + \dots \\ &= 1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \frac{\theta^6}{6!} + \dots \\ &\quad + j \left\{ \theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \dots \right\} \\ &= \end{aligned}$$

Set of column vectors:

$$\mathbb{R}^3 = \left\{ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} : x_1, x_2, x_3 \in \mathbb{R} \right\}$$

- Scalar multiplication

$$\alpha \cdot \mathbf{x} = \alpha \cdot \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} \alpha \cdot x_1 \\ \alpha \cdot x_2 \\ \alpha \cdot x_3 \end{bmatrix}, \quad \forall \alpha \in \mathbb{R}, \forall \mathbf{x} \in \mathbb{R}^3$$

- Vector addition: $\mathbf{x}, \mathbf{y} \in \mathbb{R}^3$

$$\mathbf{x} + \mathbf{y} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} x_1 + y_1 \\ x_2 + y_2 \\ x_3 + y_3 \end{bmatrix}, \quad \forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^3$$

- Commutativity of vector addition

$$\mathbf{x} + \mathbf{y} = \mathbf{y} + \mathbf{x}, \quad \forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^3$$

- Associativity of vector addition

$$\mathbf{x} + (\mathbf{y} + \mathbf{z}) = (\mathbf{x} + \mathbf{y}) + \mathbf{z}, \quad \forall \mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbb{R}^3$$

- Identity element of vector addition

$$\mathbf{0} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \implies \mathbf{x} + \mathbf{0} = \mathbf{x}, \quad \forall \mathbf{x} \in \mathbb{R}^3$$

- Inverse elements of vector addition

For each $\mathbf{x} \in \mathbb{R}^3$, $-\mathbf{x} = -1 \cdot \mathbf{x}$ satisfies $\mathbf{x} + (-\mathbf{x}) = \mathbf{0}$

- Compatibility of scalar multiplication with field multiplication

$$\alpha \cdot (\beta \cdot \mathbf{x}) = (\alpha \cdot \beta) \cdot \mathbf{x}, \quad \forall \alpha, \beta \in \mathbb{R}, \quad \forall \mathbf{x} \in \mathbb{R}^3$$

- Identity element of scalar multiplication

$$1 \cdot \mathbf{x} = \mathbf{x}, \quad \forall \mathbf{x} \in \mathbb{R}^3$$

- Distributivity of scalar multiplication w.r.t. vector addition

$$\alpha \cdot (\mathbf{x} + \mathbf{y}) = \alpha \cdot \mathbf{x} + \alpha \cdot \mathbf{y}, \quad \forall \alpha \in \mathbb{R}, \quad \forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^3$$

- Distributivity of scalar multiplication w.r.t. field addition

$$(\alpha + \beta) \cdot \mathbf{x} = \alpha \cdot \mathbf{x} + \beta \cdot \mathbf{x}, \quad \forall \alpha, \beta \in \mathbb{R}, \quad \forall \mathbf{x} \in \mathbb{R}^3$$

Vector space:

A vector space over a field $\mathbb{F}$ ($\mathbb{R}$ or $\mathbb{C}$) is a set V equipped with two binary operations (vector addition $+$: $V \times V \rightarrow V$ and scalar multiplication $\cdot$: $\mathbb{F} \times V \rightarrow V$), which satisfies the following properties for all $\alpha, \beta \in \mathbb{F}$ and for all $x, y, z \in V$.

- $x + y = y + x$
- $x + (y + z) = (x + y) + z$
- $\exists \mathbf{0} \in V$ such that $x + \mathbf{0} = x$
- For each x , there exists $-x$ such that $x + (-x) = \mathbf{0}$
- $\alpha \cdot (\beta \cdot x) = (\alpha \cdot \beta) \cdot x$
- $1 \cdot x = x$
- $\alpha \cdot (x + y) = \alpha \cdot x + \alpha \cdot y$
- $(\alpha + \beta) \cdot x = \alpha \cdot x + \beta \cdot x$

Discrete-time signals = sequences:

- A complex-valued sequence is a function from $\mathbb{Z}$ to $\mathbb{C}$.

$$\{x[n]\}_{n \in \mathbb{Z}} = \begin{bmatrix} \vdots \\ x[-1] \\ x[0] \\ x[1] \\ \vdots \end{bmatrix}$$

- Scalar multiplication

$$\alpha \cdot \{x[n]\}_{n \in \mathbb{Z}} = \alpha \cdot \begin{bmatrix} \vdots \\ x[-1] \\ x[0] \\ x[1] \\ \vdots \end{bmatrix} = \begin{bmatrix} \vdots \\ \alpha \cdot x[-1] \\ \alpha \cdot x[0] \\ \alpha \cdot x[1] \\ \vdots \end{bmatrix}$$

- Vector addition

$$\{x[n]\}_{n \in \mathbb{Z}} + \{y[n]\}_{n \in \mathbb{Z}} = \begin{bmatrix} \vdots \\ x[-1] \\ x[0] \\ x[1] \\ \vdots \end{bmatrix} + \begin{bmatrix} \vdots \\ y[-1] \\ y[0] \\ y[1] \\ \vdots \end{bmatrix} = \begin{bmatrix} \vdots \\ x[-1] + y[-1] \\ x[0] + y[0] \\ x[1] + y[1] \\ \vdots \end{bmatrix}$$

- Set of all complex-valued sequences is a vector space over scalar field $\mathbb{C}$

Alternative expressions:

- $\{x[n]\}_{n \in \mathbb{Z}}$ is a discrete-time complex-valued signal.

$\iff x$ is the label for a mapping from $\mathbb{Z}$ to $\mathbb{C}$

$\iff \mathbb{Z} \ni n \mapsto x[n] \in \mathbb{C}$

- Scalar multiplication

$$\underbrace{\begin{bmatrix} \vdots \\ y[-1] \\ y[0] \\ y[1] \\ \vdots \end{bmatrix}}_{\{y[n]\}_{n \in \mathbb{Z}}} = \alpha \cdot \underbrace{\begin{bmatrix} \vdots \\ x[-1] \\ x[0] \\ x[1] \\ \vdots \end{bmatrix}}_{\{x[n]\}_{n \in \mathbb{Z}}} = \begin{bmatrix} \vdots \\ \alpha \cdot x[-1] \\ \alpha \cdot x[0] \\ \alpha \cdot x[1] \\ \vdots \end{bmatrix}$$

$$\iff y[n] = \alpha \cdot x[n], \quad \forall n \in \mathbb{Z}$$

- Vector addition

$$\underbrace{\begin{bmatrix} \vdots \\ z[-1] \\ z[0] \\ z[1] \\ \vdots \end{bmatrix}}_{\{z[n]\}_{n \in \mathbb{Z}}} = \underbrace{\begin{bmatrix} \vdots \\ x[-1] \\ x[0] \\ x[1] \\ \vdots \end{bmatrix}}_{\{x[n]\}_{n \in \mathbb{Z}}} + \underbrace{\begin{bmatrix} \vdots \\ y[-1] \\ y[0] \\ y[1] \\ \vdots \end{bmatrix}}_{\{y[n]\}_{n \in \mathbb{Z}}} = \begin{bmatrix} \vdots \\ x[-1] + y[-1] \\ x[0] + y[0] \\ x[1] + y[1] \\ \vdots \end{bmatrix}$$

$$\iff z[n] = x[n] + y[n], \quad \forall n \in \mathbb{Z}$$

Special discrete-time signals:

- There are signals represented by a few equations.
- Impulse: $\{\delta[n]\}_{n \in \mathbb{Z}}$

$$\delta[n] = \begin{cases} 1 & n = 0, \\ 0 & n \neq 0. \end{cases}$$

- Unit-step function: $\{u[n]\}_{n \in \mathbb{Z}}$

$$u[n] = \begin{cases} 1 & n \geq 0, \\ 0 & n < 0. \end{cases}$$

- Exponential signal: $\{a^n\}_{n \in \mathbb{Z}}$
- One-sided exponential signal

$$x[n] = a^n \cdot u[n], \quad \forall n \in \mathbb{Z}$$

$$\iff \{x[n]\}_{n \in \mathbb{Z}} = \{a^n\}_{n \in \mathbb{Z}} \odot \{u[n]\}_{n \in \mathbb{Z}}$$

$$\iff x[n] = \begin{cases} a^n & n \geq 0, \\ 0 & n < 0. \end{cases}$$

Continuous-time signals = functions on $\mathbb{R}$:

- $\{x(t)\}_{t \in \mathbb{R}}$ is a continuous-time complex-valued signal.

$\iff x$ is the label for a mapping from $\mathbb{R}$ to $\mathbb{C}$

$\iff \mathbb{R} \ni t \mapsto x(t) \in \mathbb{C}$

- Scalar multiplication

$$\{y(t)\}_{t \in \mathbb{R}} = \alpha \cdot \{x(t)\}_{t \in \mathbb{R}}$$

$$\iff y(t) = \alpha \cdot x(t), \quad \forall t \in \mathbb{R}$$

- Vector addition

$$\{z(t)\}_{t \in \mathbb{R}} = \{x(t)\}_{t \in \mathbb{R}} + \{y(t)\}_{t \in \mathbb{R}}$$

$$\iff z(t) = x(t) + y(t), \quad \forall t \in \mathbb{R}$$

- Set of all complex-valued functions on $\mathbb{R}$ is a vector space over scalar field $\mathbb{C}$

Signals and systems:

- Continuous-time: $t \in \mathbb{R}$



- Discrete-time: $n \in \mathbb{Z}$



- Assumption:

All signals are *complex valued*. Below, $j \triangleq \sqrt{-1}$:

$$x[n] = \text{Re}(x[n]) + j \text{Im}(x[n]) = |x[n]| e^{j\angle x[n]}.$$

- Uniform sampling:

We say that $\{x[n]\}_{n \in \mathbb{Z}}$ is a *T-sampled version* of $\{x(t)\}_{t \in \mathbb{R}}$ when $x[n] = x(nT)$ for every integer n . Here, T denotes the sampling interval in seconds.

- Key question:

What “information” does the discrete-time sequence $\{x[n]\}_{n \in \mathbb{Z}}$ contain relative to the continuous-time waveform $\{x(t)\}_{t \in \mathbb{R}}$?

A few important system properties:

For the statements below, assume that $\{y[n]\} = \mathcal{H}\{\{x[n]\}\}$.

1. Linear:

$$\mathcal{H}\{\alpha\{x[n]\} + \beta\{w[n]\}\} = \alpha\mathcal{H}\{\{x[n]\}\} + \beta\mathcal{H}\{\{w[n]\}\}$$
$$\forall \alpha, \beta \in \mathbb{C}, \forall \{x[n]\}, \{w[n]\}$$

2. Time-invariant:

$$\mathcal{H}\{\{x[n-d]\}\} = \{y[n-d]\}, \quad \forall d \in \mathbb{Z}, \forall \{x[n]\}$$

3. Causal:

At any time m , the output value $y[m]$ does not depend on the future input values $\{x[n]\}_{n>m}$.

4. BIBO Stable:

If $\{x[n]\}$ is bounded, then $\{y[n]\}$ will be bounded.

Recall that “bounded $\{x[n]\}$ ” means that there exists some finite value M_x such that $|x[n]| < M_x$ for all n .

Note: These properties can be directly re-stated for continuous-time systems as well.

Example of system properties:

Consider the system $y(t) = |x(t)|^2$:

- Is it linear?

Linear combining at input: $|\alpha x(t) + \beta w(t)|^2$

Linear combining at output: $\alpha |x(t)|^2 + \beta |w(t)|^2$

The two are not equivalent, thus system not linear!

- Is it time-invariant?

Delay at input: $|\text{delay}_\tau\{x(t)\}|^2 = \{|x(t - \tau)|^2\}$

Delay at output: $\text{delay}_\tau\{|x(t)|^2\} = \{|x(t - \tau)|^2\}$

The two are equivalent, thus system is time-invariant.

- Is it causal?

For all t , does $y(t)$ depend only on $\{x(\tau)\}_{\tau \leq t}$?

Yes, thus system is causal.

- Is it stable?

Do all bounded inputs yield bounded outputs?

If $|x(t)| < M_x$ for all $t \in \mathbb{R}$, is there an M_y such that $|y(t)| < M_y$ for all $t \in \mathbb{R}$?

Yes: $M_y = M_x^2$ suffices. Thus system is stable.

More system examples:

1. Modulation: $y[n] = e^{j\omega_0 n} x[n]$

linear? time-invariant? causal? stable?

yes, no, yes, yes

2. Convolution: $y(t) = \int_{-\infty}^{\infty} h(\tau)x(t - \tau)d\tau$

linear? time-invariant? causal? stable?

yes, yes, depends on $h(t)$, depends on $h(t)$

3. Recursion: $y[n] = x[n] + a y[n - 1]$

linear? time-invariant? causal? stable?

$$\begin{aligned} y[n] &= x[n] + a(x[n - 1] + ay[n - 2]) \\ &= x[n] + ax[n - 1] + a^2(x[n - 2] + ay[n - 3]) \\ &= \sum_{m=0}^{\infty} a^m x[n - m] \end{aligned}$$

yes, yes, yes, for $|a| < 1$, assuming finite $|y[-\infty]|$

LTI systems — input/output behavior:

- Linear time-invariant (LTI) systems are a very important and convenient class of systems:

The input/output behavior of an LTI system is completely characterized by its impulse response!

- Discrete-time case:

- The *impulse response* of $\mathcal{H}$ is the signal

$$\{h[n]\} = \mathcal{H}\{\{\delta[n]\}\}, \text{ for Kronecker delta } \{\delta[n]\}.$$

- The *Kronecker delta* sequence $\{\delta[n]\}$ is defined as

$$\delta[n] \triangleq \begin{cases} 1 & \text{if } n = 0 \\ 0 & \text{if } n \neq 0 \end{cases},$$

and obeys the following *sifting property*:

$$y[n] = \sum_{m=-\infty}^{\infty} y[m]\delta[m-n] = \sum_{m=-\infty}^{\infty} y[m]\delta[n-m].$$

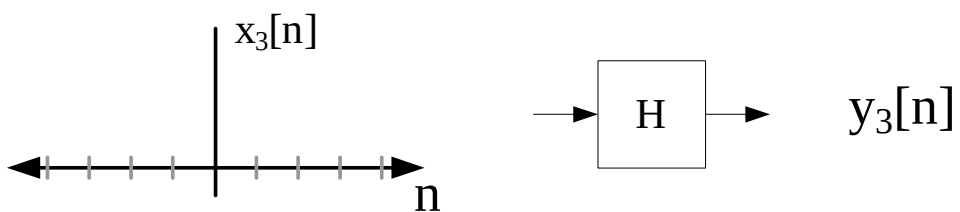
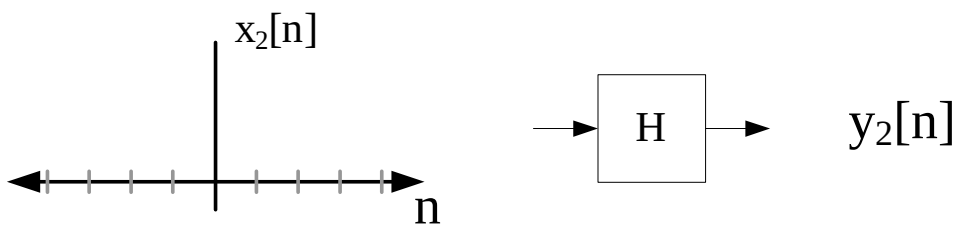
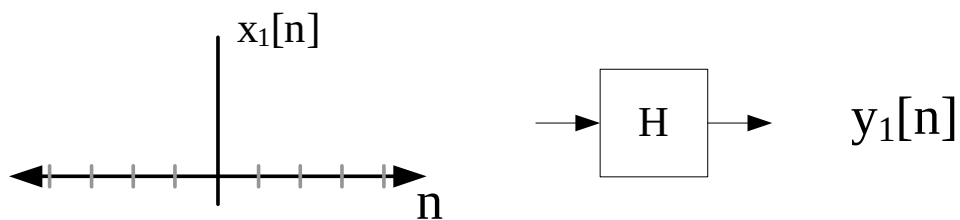
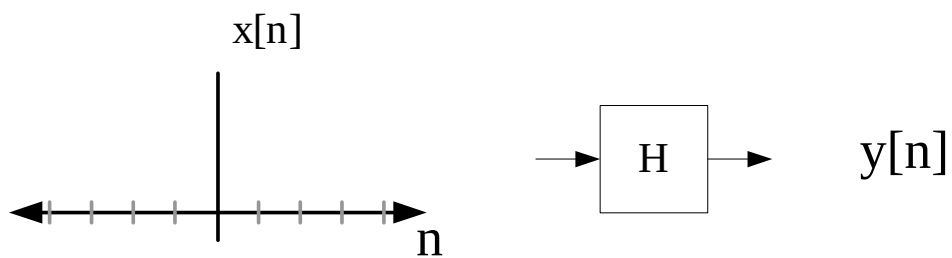
- Why does the impulse response $\{h[n]\}$ completely characterize $\mathcal{H}$?

First, write $\{x[n]\} = \sum_m x[m]\{\delta[n-m]\}$. Then

$$\begin{aligned} \mathcal{H}\{\{x[n]\}\} &= \mathcal{H}\{\sum_m x[m]\{\delta[n-m]\}\} \\ &= \sum_m x[m]\mathcal{H}\{\{\delta[n-m]\}\} \quad (\text{linearity}) \\ &= \sum_m x[m]\{h[n-m]\} \quad (\text{time-invariance}). \end{aligned}$$

- Thus, the system's output equals the *convolution* of the input $\{x[n]\}$ with the impulse response $\{h[n]\}$!

Example:

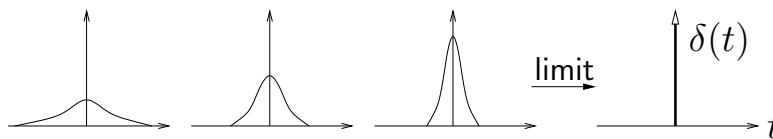


- Continuous-time case:

- The *impulse response* of $\mathcal{H}_c$ is the signal

$$\{h(t)\} = \mathcal{H}_c\{\{\delta(t)\}\}, \text{ for Dirac delta } \{\delta(t)\}.$$

- The *Dirac delta* waveform can be understood as the limit of increasingly skinny *unit-area* pulses:



and it obeys the following *sifting property*:

$$y(t) = \int_{-\infty}^{\infty} y(\tau)\delta(\tau - t)d\tau = \int_{-\infty}^{\infty} y(\tau)\delta(t - \tau)d\tau.$$

- Why does the impulse response $\{h(t)\}$ completely characterize $\mathcal{H}_c$?

First, write $\{x(t)\} = \int x(\tau)\{\delta(t - \tau)\}d\tau$. Then

$$\begin{aligned} \mathcal{H}_c\{\{x(t)\}\} &= \mathcal{H}_c\left\{\int x(\tau)\{\delta(t - \tau)\}d\tau\right\} \\ &= \int x(\tau)\mathcal{H}_c\{\{\delta(t - \tau)\}\}d\tau \quad (\text{linearity}) \\ &= \int x(\tau)\{h(t - \tau)\}d\tau \quad (\text{time-invariance}). \end{aligned}$$

- Thus, the system's output equals the *convolution* of the input $\{x(t)\}$ with the impulse response $\{h(t)\}$!

- We usually denote the convolution operator by “*”:

$$\begin{aligned} \int x(\tau)\{h(t - \tau)\}d\tau &= \{x(t)\} * \{h(t)\} \\ \sum_m x[m]\{h[n - m]\} &= \{x[n]\} * \{h[n]\} \end{aligned}$$

Computing Convolution:

1. Let the computer do it.

In Matlab,

```
>> y = conv(h,x);
```

Example:

$$x[n] = u[n] - u[n - 5]$$

$$h[n] = 2\delta[n] - \delta[n - 1] + 3\delta[n - 2]$$

```
>> x = [1, 1, 1, 1, 1];
```

```
>> h = [2, -1, 3];
```

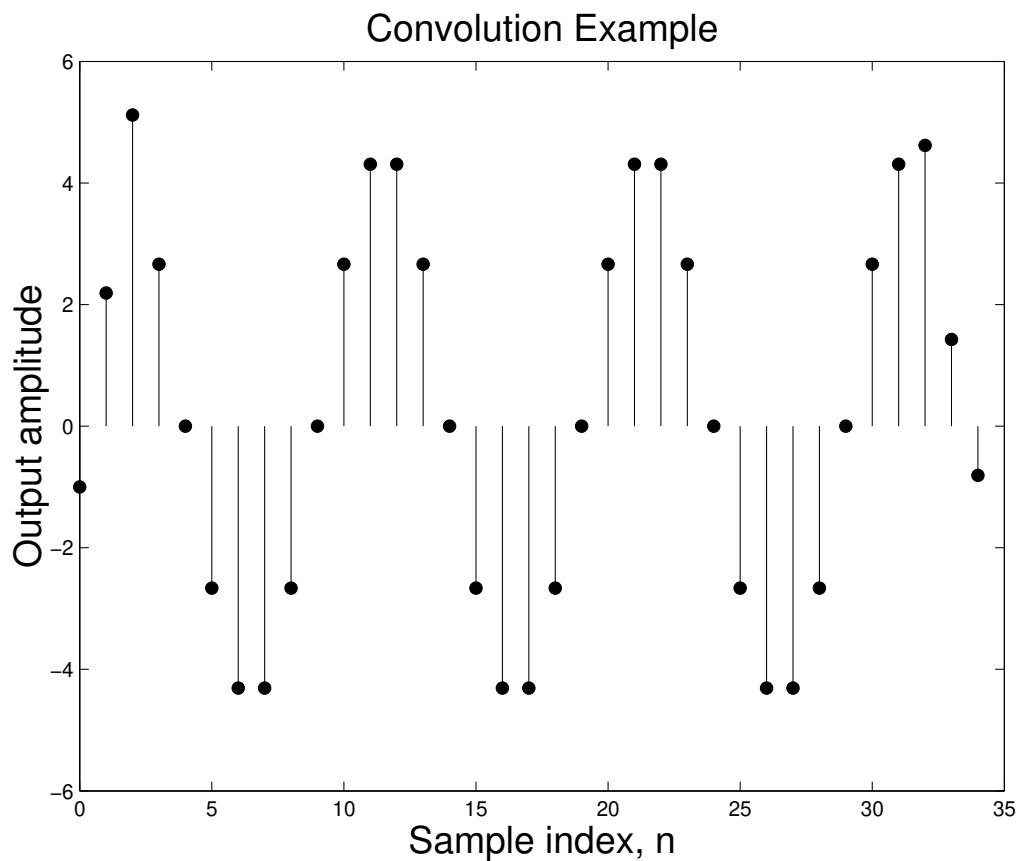
```
>> y = conv(x,h)
```

y =

2 1 4 4 4 2 3

Matlab Example:

```
>> n = [0:1:31];  
>> x = cos(0.2*pi*n);% cosine, amplitude 1, 0.2*pi radians  
>> h = [-1 3 3 -1];% impulse response  
>> y = conv(x,h)  
>> % construct a plot  
>> stem([0:length(y)-1], y,'filled');
```



Computing Convolution:

2. Graphical convolution.

“Flip, shift, multiply, add.”

Very useful for a fast, qualitative, visual interpretation.

Discrete convolution demo in Matlab: <http://dspfirst.gatech.edu/matlab/>

Computing Convolution:

3. Matrix-vector multiply (for finite duration signals)

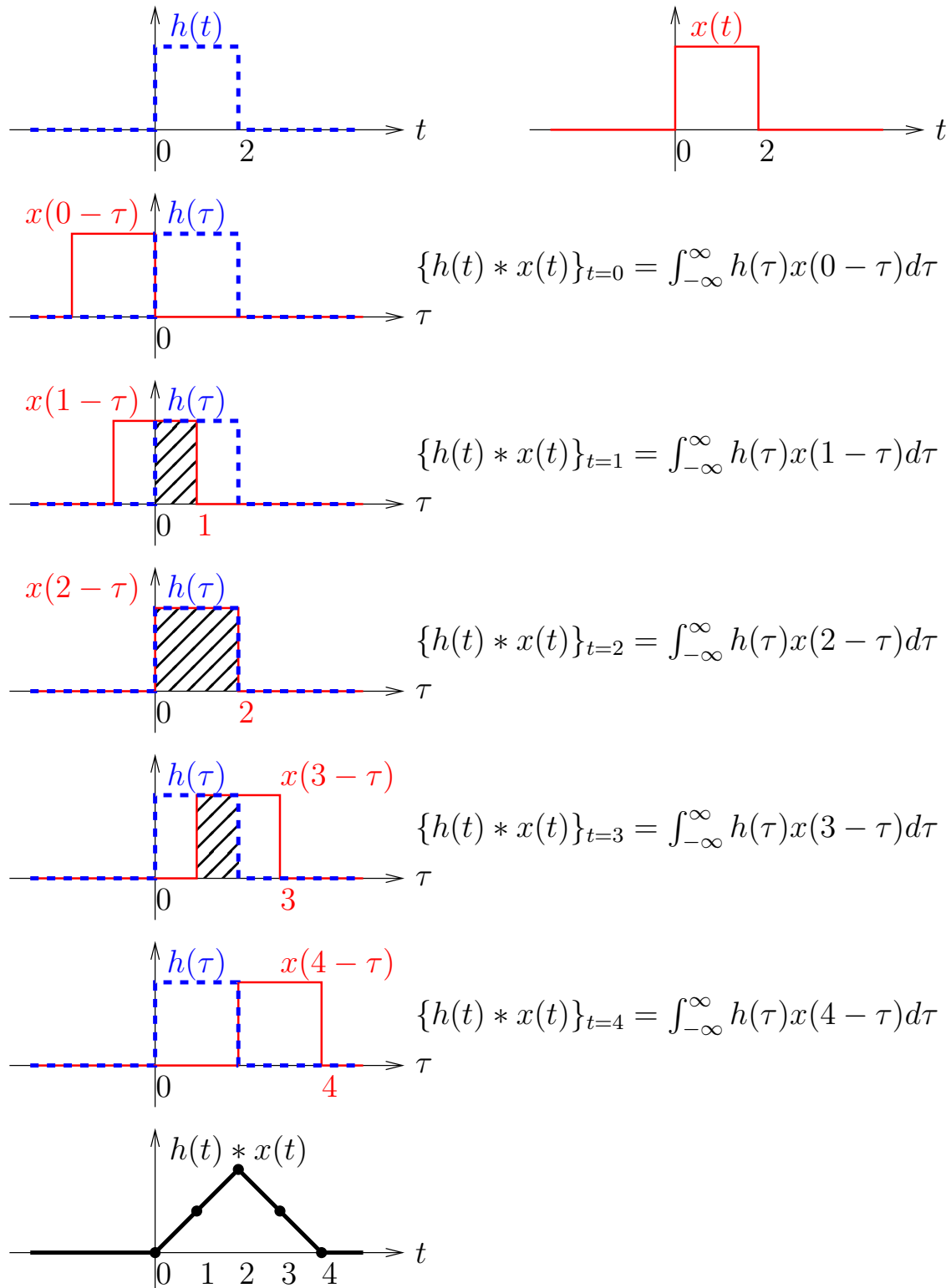
$$y[n] = \sum_k x[k] h[n - k] = \sum_k h[k] x[n - k]$$

Example: $M = 3$; $x[n]$ zero for $n < 0$ and $n > 7$.

$$\begin{bmatrix} x[0] & 0 & 0 & 0 \\ x[1] & x[0] & 0 & 0 \\ x[2] & x[1] & x[0] & 0 \\ x[3] & x[2] & x[1] & x[0] \\ x[4] & x[3] & x[2] & x[1] \\ x[5] & x[4] & x[3] & x[2] \\ x[6] & x[5] & x[4] & x[3] \\ x[7] & x[6] & x[5] & x[4] \\ 0 & x[7] & x[6] & x[5] \\ 0 & 0 & x[7] & x[6] \\ 0 & 0 & 0 & x[7] \end{bmatrix} \begin{bmatrix} h[0] \\ h[1] \\ h[2] \\ h[3] \end{bmatrix} = \begin{bmatrix} y[0] \\ y[1] \\ y[2] \\ y[3] \\ y[4] \\ y[5] \\ y[6] \\ y[7] \\ y[8] \\ y[9] \\ y[10] \end{bmatrix}$$

CT Convolution graphically:

Graphically, convolution can be understood as “flip and slide”:



LTI systems — convolution:

Some important properties of convolution:

1. Commutative: $x * h = h * x$

[[block diagrams](#)]

2. Associative: $(x * h) * w = x * (h * w)$

[[block diagrams](#)]

3. Distributive: $x * (h + w) = x * w + x * h$

[[block diagrams](#)]

4. Identity: $h * \delta = h$

[[block diagrams](#)]

These apply in both discrete-time and continuous-time cases.

LTI systems — causality:

- An LTI system is *causal* iff $h[n] = 0$ for all $n < 0$
- Discrete-time case:
 $\Leftarrow$ (Assume $h[n] = 0$ for all $n < 0$ and show causal.)
For an LTI system, we know

$$\begin{aligned}y[m] &= \sum_{n=-\infty}^{\infty} h[n]x[m-n] \\ &= \sum_{n=0}^{\infty} h[n]x[m-n] \text{ (since } h[n]|_{n<0} = 0),\end{aligned}$$

so $y[m]$ depends only on $x[m], x[m-1], x[m-2], \dots$

Thus, system is causal.

- $\Rightarrow$ (Assume causal and show $h[n] = 0$ for all $n < 0$, or assume $h[n] \neq 0$ for some $n < 0$ and show noncausal.)

For an LTI system,

$$y[m] = \sum_{n=-\infty}^{\infty} h[n]x[m-n].$$

Assuming $h[-d] \neq 0$ for some $d > 0$, can see that

$y[m]$ depends on the future value $x[m+d]$.

Thus, the system is noncausal.

- Continuous-time case: same idea...

LTI systems — stability:

- An LTI system is (BIBO) *stable* iff its impulse response is *absolutely summable* ($\sum_n |h[n]| < \infty$ or $\int |h(t)|dt < \infty$).

Can we prove this?

- Discrete-time case:

$\Leftarrow$ (Assume $\sum_n |h[n]| < \infty$ and show system is stable.)

For an LTI system, know

$$\begin{aligned} |y[n]| &= \left| \sum_m x[m]h[n-m] \right| \\ &\leq \sum_m |x[m]h[n-m]| \\ &= \sum_m |x[m]| |h[n-m]|. \end{aligned}$$

and if input is bounded (i.e., $|x[m]| < M_x$ for all m),

$$< M_x \underbrace{\sum_m |h[n-m]|}_{< \infty},$$

so the output is also bounded.

$\Rightarrow$ (Assume system is stable and show $\sum_n |h[n]| < \infty$, or assume $\sum_n |h[n]| = \infty$ and show system is unstable.)

Consider the bounded input $x[m] = \text{sgn}(h[-m])$:

$$\begin{aligned} y[n] &= \sum_m \text{sgn}(h[-m])h[n-m] \\ y[0] &= \sum_m |h[-m]| = \infty. \end{aligned}$$

Since the output is unbounded, the system is unstable.

- Continuous-time case: same idea...

An algebra for LTI systems?:

- We've seen that an LTI system's output can be written as the convolution of the input and the impulse response:

$$\mathcal{H}\{\{x[n]\}\} = \sum_{m=-\infty}^{\infty} x[m]\{h[n-m]\}$$
$$\mathcal{H}_c\{\{x(t)\}\} = \int_{-\infty}^{\infty} x(\tau)\{h(t-\tau)\}d\tau.$$

- But integrals and infinite-sums are a pain! Is there an “*algebra*” for signals and systems? (Algebra involves only multiplications & additions.)

Yes! Through the “frequency domain”...

- We first review the details for continuous-time (CT) systems, and later proceed to discrete-time (DT) systems.
- As we shall see, there are many forms of the “*frequency domain*”, including [Fourier series](#), the [CTFT](#), the [Laplace transform](#), the [DTFT](#), the [DFT](#), and the [Z-transform](#). Understanding the relationships among these different “frequency” representations is one of the primary objectives of this course.

Fourier Series:

- A periodic signal, with period P , satisfies

$$x(t) = x(t + nP) \text{ for all integers } n.$$

Examples: sine wave, square wave, triangle wave, ...

- The Fourier series (FS) says that a P -periodic signal $x(t)$ can be written as an infinite sum of complex exponentials:

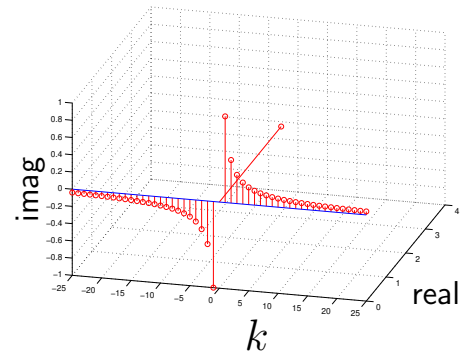
$$x(t) = \sum_{k=-\infty}^{\infty} x_k e^{j\frac{2\pi}{P}kt}, \quad x_k \triangleq \frac{1}{P} \int_0^P x(t) e^{-j\frac{2\pi}{P}kt} dt,$$

where $\{x_k\}_{k \in \mathbb{Z}}$ are known as the FS coefficients.

- Example — sawtooth wave:

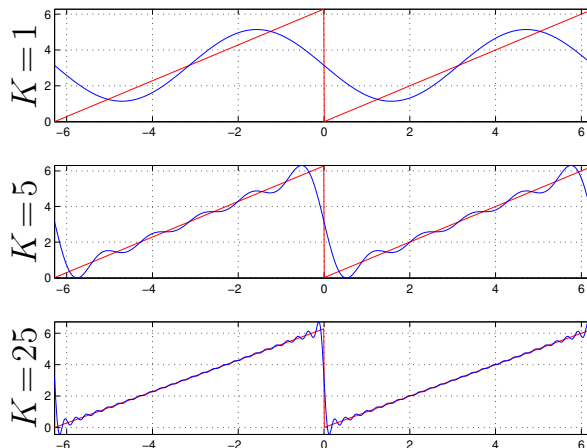
$$x(t) = t \text{ for } t \in [0, P), P=2\pi$$

$$x_k = \frac{j}{k} \big|_{k \neq 0}, \quad x_0 = \frac{P}{2}$$



K -truncated FS

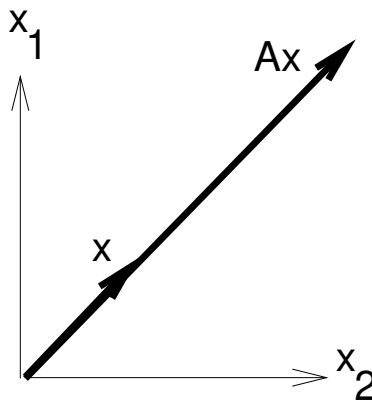
$$\sum_{k=-K}^K x_k e^{j\frac{2\pi}{P}kt}$$



Eigenfunctions of DT LTI Systems:

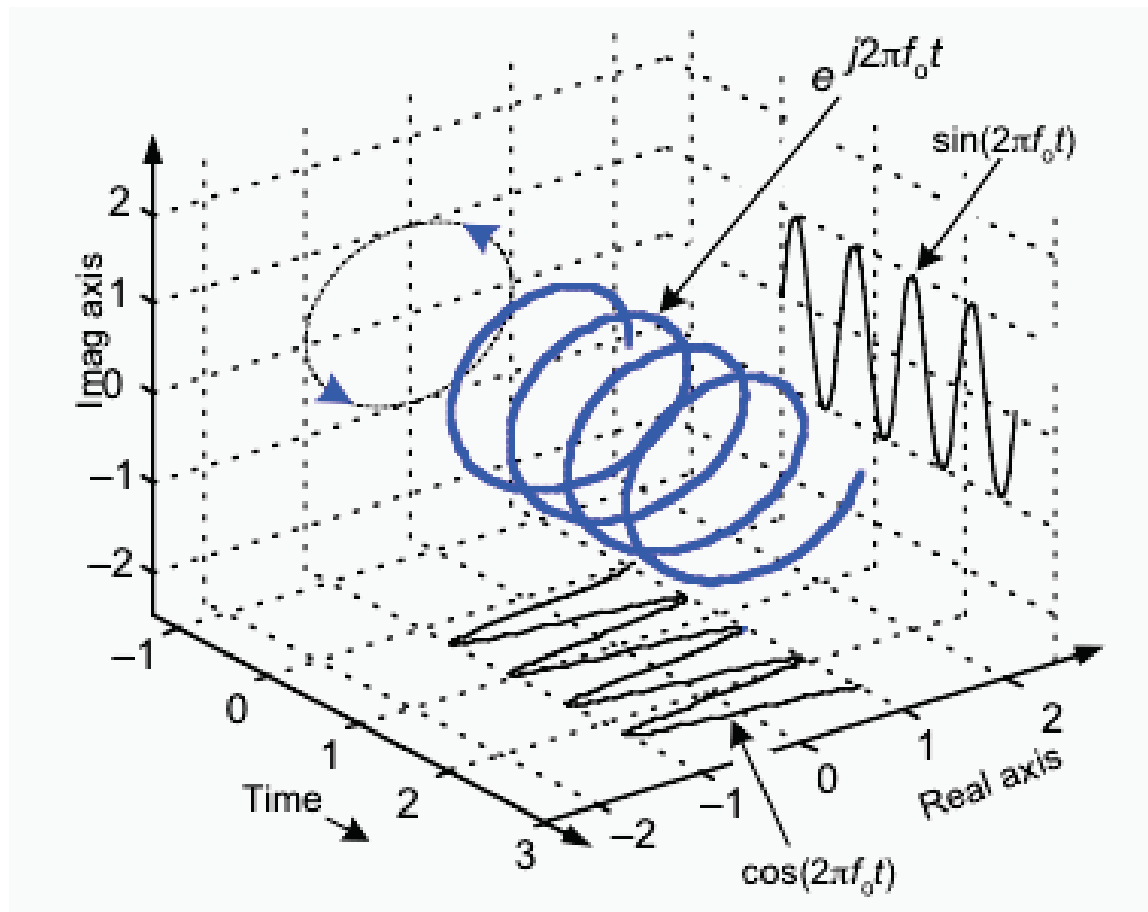
- An analogy:

$$A\mathbf{x} = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \end{bmatrix} = 4 \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$



- LTI System:

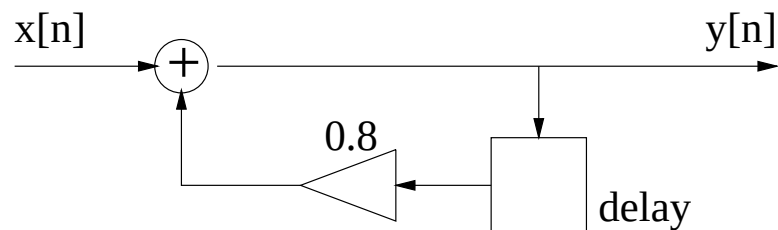
$$\{e^{j\omega n}\}_{n \in \mathbb{Z}} \longrightarrow \boxed{\mathcal{H}} \longrightarrow \lambda_{\omega} \cdot \{e^{j\omega n}\}_{n \in \mathbb{Z}}$$



IIR Example:

Consider a recursive LTI system:

$$y[n] = x[n] + 0.8y[n - 1]$$



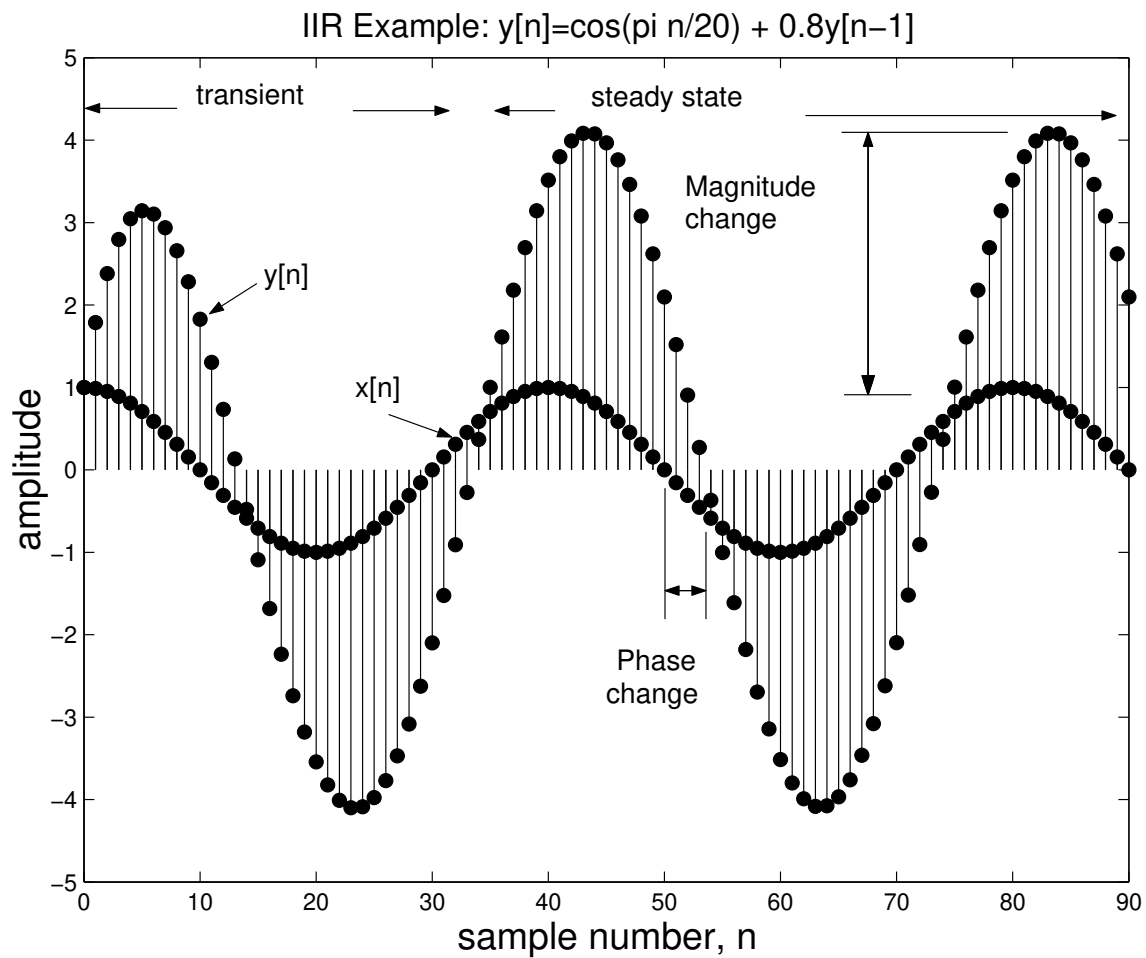
$$x[n] = \cos(\pi n/20)u[n]$$

$$|H(e^{j\omega})| = 4.093 \quad \text{for } \omega = \pi/20$$

$$\angle H(e^{j\omega}) = -0.538 \text{ radians} \quad \text{for } \omega = \pi/20$$

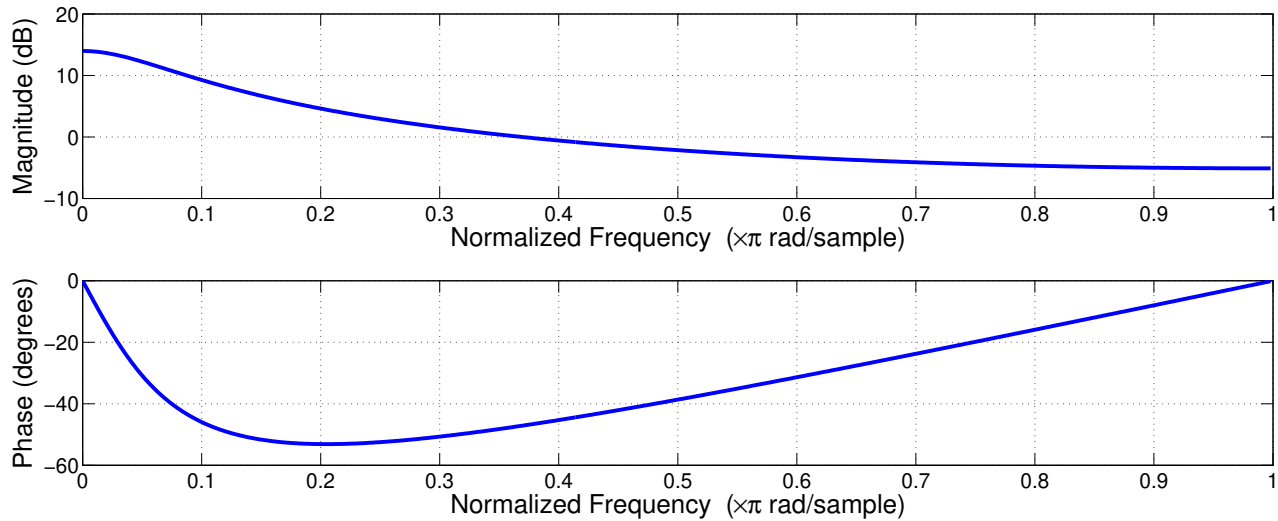
$$y_{ss}[n] = 4.093 \cos(\pi n/20 - 0.538)$$

IIR Example, continued:



IIR Example, continued:

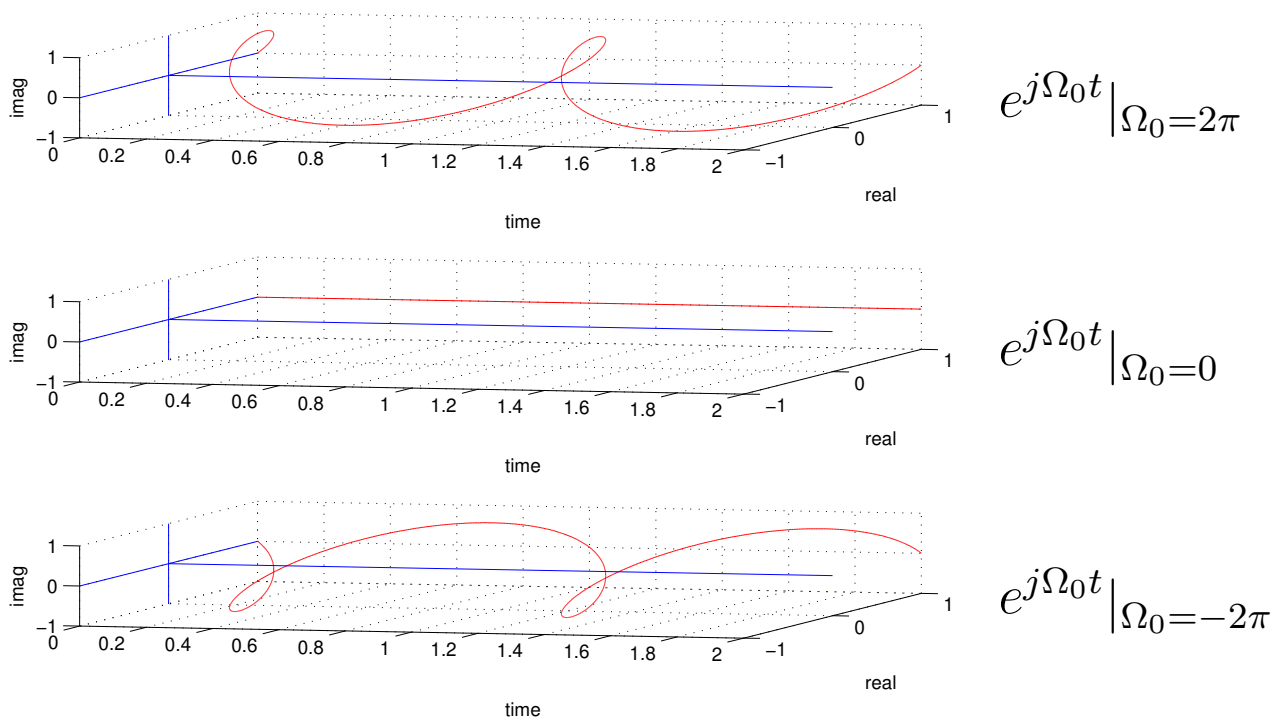
Frequency response plots (magnitude and phase)



```
>> freqz([1],[1 -0.8])
```

Negative frequencies:

- The CTFT allows negative frequencies. What are they?
- Recall that a “pure” tone at Ω_0 radians/sec can be written as $e^{j\Omega_0 t} \xleftrightarrow{\text{CTFT}} 2\pi\delta(\Omega - \Omega_0)$.
- The signal $e^{j\Omega_0 t}$ is a corkscrew: for $\Omega_0 > 0$, it spins counter-clockwise and, for $\Omega_0 < 0$, it spins clockwise.



- In essence, the CTFT implies that any signal can be constructed as a sum of (possibly infinitely many) scaled corkscrews, each with a distinct frequency. The sign of the frequency simply denotes the direction of spin.

- What if the signal is a real-valued sinusoid?

$$\begin{aligned}\cos(\Omega_0 t + \theta) &= \frac{1}{2} \left(e^{j(\Omega_0 t + \theta)} + e^{-j(\Omega_0 t + \theta)} \right) \\ &= \frac{e^{j\theta}}{2} e^{j\Omega_0 t} + \frac{e^{-j\theta}}{2} e^{-j\Omega_0 t}\end{aligned}$$

Thus it contains both positive and negative frequencies.

- How do we generate/represent complex signals in hardware?

$$e^{j(\Omega_0 t + \theta)} = \cos(\Omega_0 t + \theta) + j \sin(\Omega_0 t + \theta)$$

Using two real-valued wires/buses.

FIR Example:

$$y[n] = 0.5x[n] + x[n - 1] + 0.5x[n - 2]$$

This is a second-order FIR filter, and hence is a linear time-invariant system. We have that

$$h[n] = 0.5\delta[n] + \delta[n - 1] + 0.5\delta[n - 2]$$

To find $H(e^{j\omega})$, we start with the definition

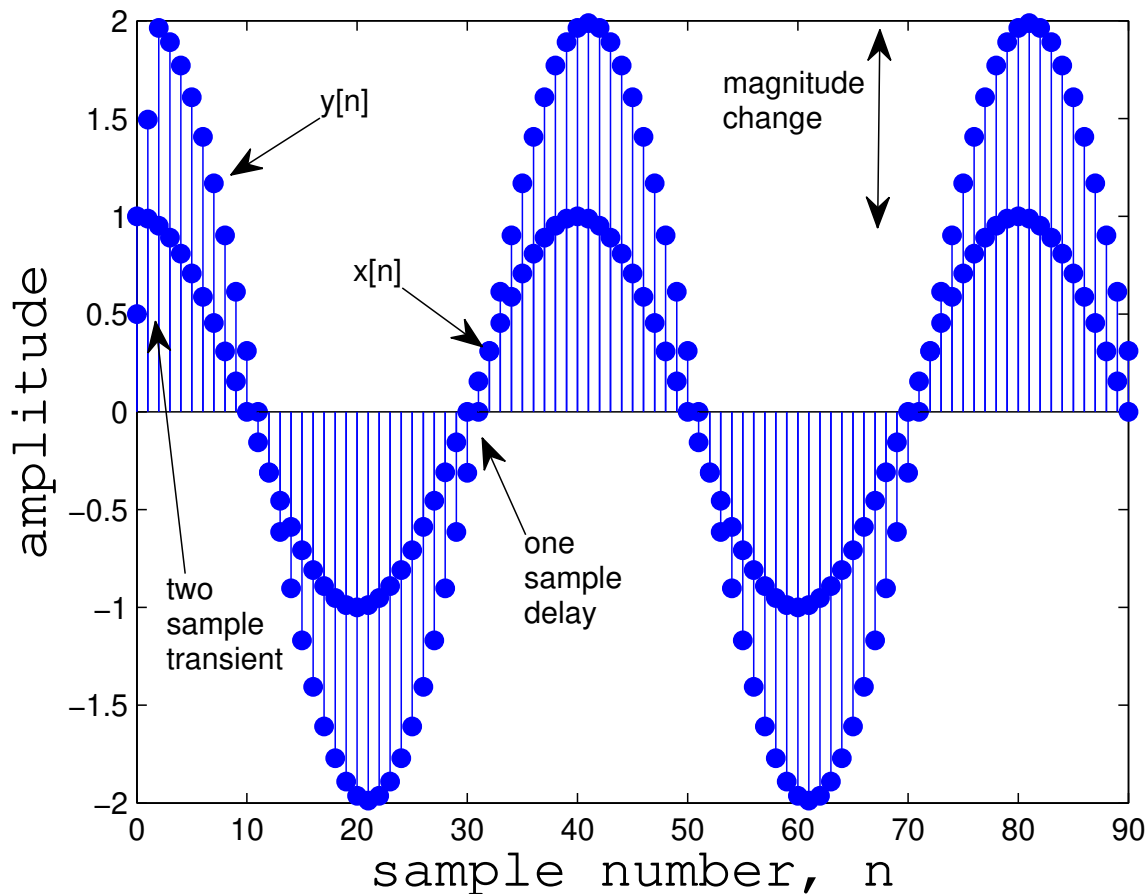
$$\begin{aligned} H(e^{j\omega}) &= \sum_{k=-\infty}^{\infty} h[k]e^{-j\omega k} \\ &= \frac{1}{2}e^0 + 1e^{-j\omega} + \frac{1}{2}e^{-j\omega 2} \\ &= e^{-j\omega} \left\{ \frac{1}{2}e^{j\omega} + 1 + \frac{1}{2}e^{-j\omega} \right\} \\ &= e^{-j\omega} \{1 + \cos(\omega)\} \end{aligned}$$

Thus, $|H(e^{j\omega})| = 1 + \cos(\omega)$ and $\angle H(e^{j\omega}) = -\omega$.

FIR Example, continued:

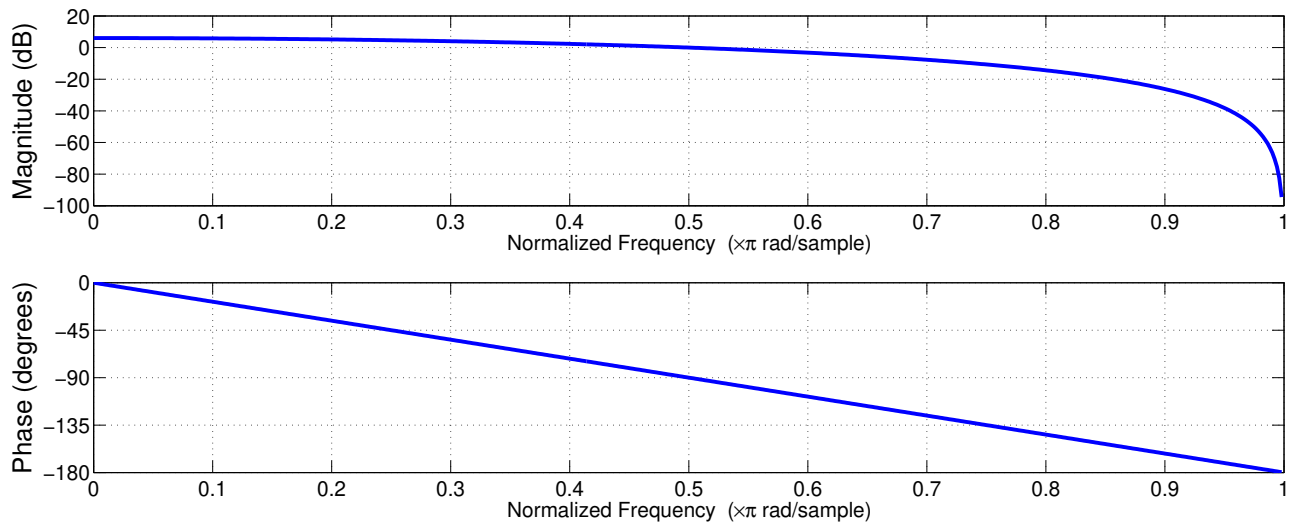
For $x[n] = \cos(\pi n/20)u[n]$ we have $A = 1$, $\omega = \pi/20$ and $\phi = 0$. Thus,

$$\begin{aligned}y_{ss}[n] &= A |H(e^{j\omega})| \cos(\omega n + \phi + \angle H(e^{j\omega})) \\&= 1 \cdot |H(e^{j\pi/20})| \cos(\pi n/20 + 0 + \angle H(e^{j\pi/20})) \\&= 1.988 \cos(\pi n/20 - \pi/20)\end{aligned}$$



FIR Example, continued:

Frequency response plots (magnitude and phase)



```
>> b = [0.5, 1, 0.5];  
>> a=1;  
>> freqz(b,a);
```

Eigensignals of CT-LTI systems:

- Consider complex *sinusoidal* input $x(t) = e^{j\Omega_0 t}$, $\Omega_0 \in \mathbb{R}$.

For LTI system $\mathcal{H}_c$, we have

$$\begin{aligned}\mathcal{H}_c\{x(t)\} &= \int h(\tau)x(t-\tau)d\tau = \int h(\tau)e^{j\Omega_0(t-\tau)}d\tau \\ &= e^{j\Omega_0 t} \underbrace{\int h(\tau)e^{-j\Omega_0 \tau}d\tau}_{\triangleq H_c(j\Omega_0)} = x(t) \underbrace{H_c(j\Omega_0)}_{\text{scaling factor}},\end{aligned}$$

using Laplace-transform notation (more on this later...).

The output is simply a scaled version of the input!

- A signal which passes through system $\mathcal{H}_c$ with only a gain/phase change is an “*eigensignal*” of $\mathcal{H}_c$, and the corresponding scaling factor is its “*eigenvalue*.”

Sinusoids are eigensignals of LTI systems.

- This property yields an important implication for any *P-periodic signal* $x(t)$. Notice that if we expand such a signal via its *Fourier series* (FS) as

$$x(t) = \sum_{k=-\infty}^{\infty} x_k e^{j\frac{2\pi}{P}kt}$$

then the linearity of $\mathcal{H}_c$ implies

$$\mathcal{H}_c\{x(t)\} = \sum_{k=-\infty}^{\infty} \underbrace{x_k \overbrace{H_c(j\frac{2\pi k}{P})}^{h_k}}_{y_k} e^{j\frac{2\pi}{P}kt}.$$

- Thus, if we represent the input signal by $\{x_k\}_{k \in \mathbb{Z}}$, the output signal by $\{y_k\}_{k \in \mathbb{Z}}$, and the system by $\{h_k\}_{k \in \mathbb{Z}}$, then the operation of passing the signal through the system is simply multiplication of the corresponding coefficients: $y_k = h_k x_k$!

For an LTI system with a periodic input, the output can be computed algebraically.

- Note, however, that the example above applies only to *periodic* signals. Can we extend it to non-periodic signals?

Yes, through the continuous-time Fourier transform (CTFT)...

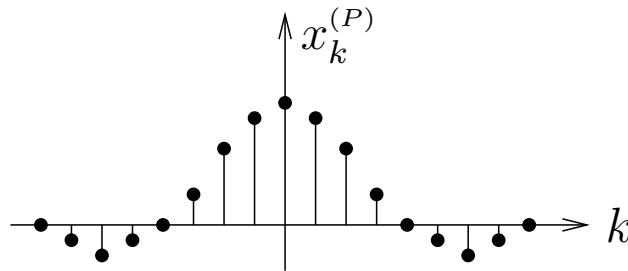
The CTFT — Derivation:

- We now derive the *continuous-time Fourier transform* (CTFT) as the $P \rightarrow \infty$ limit of the Fourier series (FS).

- Consider an arbitrary P -periodic signal $x^{(P)}(t)$ with corresponding FS coefficients $\{x_k^{(P)}\}_{k \in \mathbb{Z}}$, where

$$x_k^{(P)} = \frac{1}{P} \int_0^P x^{(P)}(t) e^{-j \frac{2\pi}{P} kt} dt = \frac{1}{P} \int_{-P/2}^{P/2} x^{(P)}(t) e^{-j \frac{2\pi}{P} kt} dt.$$

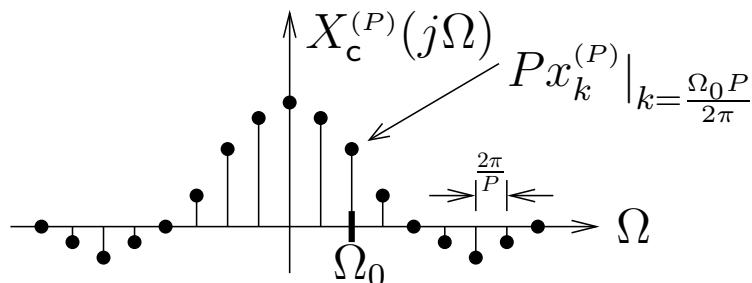
- The collection $\{x_k^{(P)}\}_{k \in \mathbb{Z}}$ constitutes the FS “spectrum”:



- Now define the “frequency” $\Omega \triangleq \frac{2\pi k}{P}$ (in rad/sec) and, from it, the (scaled) spectrum

$$X_c^{(P)}(j\Omega) \triangleq P x_k^{(P)} = \int_{-P/2}^{P/2} x^{(P)}(t) e^{-j\Omega t} dt.$$

- We can visualize the spectrum $X_c^{(P)}(j\Omega)$ as follows:



- Using FS, we can relate $X_c^{(P)}(j\Omega)$ to $x^{(P)}(t)$ as follows:

$$x^{(P)}(t) = \sum_{k=-\infty}^{\infty} x_k^{(P)} e^{j\frac{2\pi}{P}tk} = \frac{1}{2\pi} \sum_{k=-\infty}^{\infty} X_c^{(P)}(j\Omega) e^{j\Omega t \frac{2\pi}{P}}$$

- Key idea: as P increases, the # of spectral lines per Hz in $X_c^{(P)}(j\Omega)$ increases, so that $X_c^{(P)}(j\Omega)$ approaches a continuum! In this case, Riemann sum arguments allow us to write the infinite sum as an integral. So, for $x(t) \triangleq \lim_{P \rightarrow \infty} x^{(P)}(t)$, which is no longer periodic, and $X_c(j\Omega) \triangleq \lim_{P \rightarrow \infty} X_c^{(P)}(j\Omega)$, we get

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X_c(j\Omega) e^{j\Omega t} d\Omega$$

$$X_c(j\Omega) = \int_{-\infty}^{\infty} x(t) e^{-j\Omega t} dt,$$

known as the CTFT.

- The convolution operation $y(t) = \int h(\tau) x(t - \tau) d\tau$ becomes algebraic under the CTFT:

$$\begin{aligned} Y_c(j\Omega) &= \int \left[\int h(\tau) x(t - \tau) d\tau \right] e^{-j\Omega t} dt \\ &= \int \int h(\tau) e^{-j\Omega\tau} x(t - \tau) e^{-j\Omega(t-\tau)} dt d\tau \\ &= \int h(\tau) e^{-j\Omega\tau} d\tau \int x(\rho) e^{-j\Omega\rho} d\rho \\ &= H_c(j\Omega) X_c(j\Omega). \end{aligned}$$

The CTFT — important transform pairs:

- The CTFT: $X_c(j\Omega) = \int_{-\infty}^{\infty} x(t)e^{-j\Omega t} dt$ for $\Omega \in \mathbb{R}$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X_c(j\Omega)e^{j\Omega t} d\Omega,$$

is often written in “transform pair” notation

$$x(t) \xleftrightarrow{\text{CTFT}} X_c(j\Omega).$$

- Some important examples of CTFT transform pairs are:

- Dirac delta:

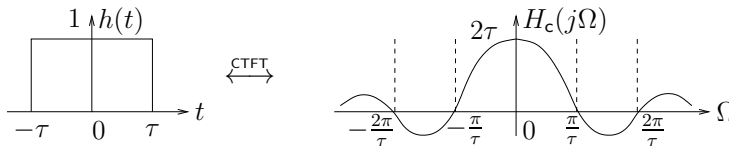
$$\delta(t - \tau) \xleftrightarrow{\text{CTFT}} e^{-j\Omega\tau}$$

- Complex exponential:

$$e^{j\Omega_0 t} \xleftrightarrow{\text{CTFT}} 2\pi\delta(\Omega - \Omega_0)$$

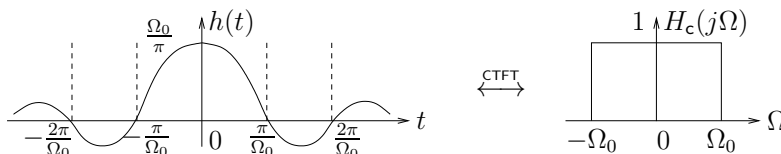
- Rectangle:

$$\begin{cases} 1 & \text{if } |t| < \tau \\ 0 & \text{else} \end{cases} \xleftrightarrow{\text{CTFT}} 2\frac{\sin(\Omega\tau)}{\Omega}$$



- Sinc:

$$\frac{\sin(\Omega_0 t)}{\pi t} \xleftrightarrow{\text{CTFT}} \begin{cases} 1 & \text{if } |\Omega| < \Omega_0 \\ 0 & \text{else} \end{cases}$$



The CTFT — derivations:

- Gaussian pulse

$$\frac{1}{\sigma\sqrt{2\pi}} \cdot e^{-\frac{t^2}{2\sigma^2}} \xleftrightarrow{\text{CTFT}} e^{-\frac{\sigma^2\Omega^2}{2}}$$

- Impulse

$$\lim_{\sigma \rightarrow 0} \frac{1}{\sigma\sqrt{2\pi}} \cdot e^{-\frac{t^2}{2\sigma^2}} \xleftrightarrow{\text{CTFT}} \lim_{\sigma \rightarrow 0} e^{-\frac{\sigma^2\Omega^2}{2}}$$

- Constant

$$\lim_{\sigma \rightarrow \infty} e^{-\frac{t^2}{2\sigma^2}} \xleftrightarrow{\text{CTFT}} \lim_{\sigma \rightarrow \infty} 2\pi \cdot \frac{\sigma}{\sqrt{2\pi}} \cdot e^{-\frac{\sigma^2\Omega^2}{2}}$$

The CTFT — important properties:

Below, assume $x(t) \xleftrightarrow{\text{CTFT}} X_c(j\Omega)$ and $w(t) \xleftrightarrow{\text{CTFT}} W_c(j\Omega)$.

1. Linearity:

$$\alpha x(t) + \beta w(t) \xleftrightarrow{\text{CTFT}} \alpha X_c(j\Omega) + \beta W_c(j\Omega)$$

2. Conjugate symmetry:

$$x^*(t) = x(-t) \Leftrightarrow X_c(j\Omega) \in \mathbb{R}$$

$$x(t) \in \mathbb{R} \Leftrightarrow X_c^*(j\Omega) = X_c(-j\Omega)$$

3. Shift/modulation:

$$x(t - \tau) \xleftrightarrow{\text{CTFT}} e^{-j\Omega\tau} X_c(j\Omega)$$

$$e^{j\Omega_0 t} x(t) \xleftrightarrow{\text{CTFT}} X_c(j(\Omega - \Omega_0))$$

4. Reversal:

$$x(-t) \xleftrightarrow{\text{CTFT}} X_c(-j\Omega)$$

5. Multiplication/convolution:

$$x(t)w(t) \xleftrightarrow{\text{CTFT}} \frac{1}{2\pi} \int_{-\infty}^{\infty} X_c(j\theta) W_c(j(\Omega - \theta)) d\theta$$

$$\int_{-\infty}^{\infty} x(\tau) w(t - \tau) d\tau \xleftrightarrow{\text{CTFT}} X_c(j\Omega) W_c(j\Omega)$$

6. Energy conservation (“Parseval’s theorem”):

$$\int_{-\infty}^{\infty} |x(t)|^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} |X_c(j\Omega)|^2 d\Omega$$

The CTFT — integration property:

- CTFT of sign function

$$\text{sgn}(t) = \begin{cases} 1 & t > 0 \\ 0 & t = 0 \\ -1 & t < 0 \end{cases} \xleftrightarrow{\text{CTFT}} \frac{2}{j\Omega}$$

Proof: Let $x_a(t) = e^{-at}u(t) - e^{at}u(-t)$ for $a > 0$. Since

$$e^{-at}u(t) \xleftrightarrow{\text{CTFT}} \frac{1}{a + j\Omega}, \quad e^{at}u(-t) \xleftrightarrow{\text{CTFT}} \frac{1}{a - j\Omega},$$

we have

$$\text{sgn}(t) = \lim_{a \rightarrow 0} x_a(t) \xleftrightarrow{\text{CTFT}} \lim_{a \rightarrow 0} \left(\frac{1}{a + j\Omega} - \frac{1}{a - j\Omega} \right).$$

- CTFT of step function

$$u(t) \xleftrightarrow{\text{CTFT}} \frac{1}{j\Omega} + \pi\delta(\Omega)$$

Proof: Note that $u(t) = \frac{\text{sgn}(t) + 1}{2}$. Recall

$$\text{sgn}(t) \xleftrightarrow{\text{CTFT}} \frac{2}{j\Omega}, \quad 1 \xleftrightarrow{\text{CTFT}} 2\pi\delta(\Omega).$$

- Integration property

$$\int_{-\infty}^t x(\lambda) d\lambda \xleftrightarrow{\text{CTFT}} \frac{X_c(j\Omega)}{j\Omega} + \pi X_c(j0) \delta(\Omega)$$

Proof: Note

$$\int_{-\infty}^t x(\lambda) d\lambda = \int_{-\infty}^{\infty} x(\lambda) u(t - \lambda) d\lambda = x(t) * u(t).$$

Due to the convolution property, we have

$$x(t) * u(t) \xleftrightarrow{\text{CTFT}} X_c(j\Omega) \left(\frac{1}{j\Omega} + \pi \delta(\Omega) \right).$$

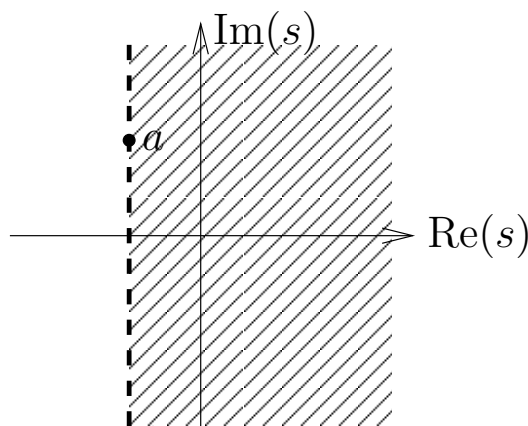
The Laplace transform:

- Recall the 2-sided *Laplace transform* (LT) of $x(t)$:

$$X_c(s) = \int_{-\infty}^{\infty} x(t)e^{-st}dt, \text{ for } s \in \mathbb{C} \text{ such that } |X_c(s)| < \infty.$$

- The CTFT can be interpreted as a special case of the LT for which $s = j\Omega$ with $\Omega \in \mathbb{R}$, and that is why we write the CTFT of $x(t)$ as “ $X_c(j\Omega)$.”
- The LT of a given $x(t)$ is defined only for those $s \in \mathbb{C}$ such that $|X_c(s)| < \infty$. This portion of the complex plane is known as the “*region of convergence* (ROC) of $X_c(s)$.”

For example, if $x(t) = e^{at}u(t)$, where $a \in \mathbb{C}$ and $u(t)$ is the Heaviside step function ($u(t) = 1$ for $t \geq 0$ and $u(t) = 0$ otherwise), then one can show that the ROC is the set $\{s \in \mathbb{C} : \text{Re}(s) > \text{Re}(a)\}$.



- Like the CTFT, the utility of the LT stems from the fact that the convolution integral $y(t) = \int h(\tau)x(t - \tau)dt$ simplifies to the algebraic expression $Y_c(s) = H_c(s)X_c(s)$.
- For many signals $x(t)$ (e.g., impulse, step, ramp, decaying exponentials), there exist convenient closed-form expressions for $X_c(s)$. Moreover, when $Y_c(s)$ is a ratio of polynomials, its inverse LT can be calculated using *partial fraction expansion* (PFE) and table lookup.
- The *advantage* of the LT vs. the CTFT is that there is a much larger class of signals with closed-form transforms. The *disadvantage* of the LT vs. the CTFT is that the inverse transform step (requiring PFE or contour integration) is more complicated.

In practical DSP, the FT is computed numerically and so we don't care too much about closed-form transforms.

Thus, in this course, we focus on the simpler CTFT.

Stability of LTI systems:

An LTI system is stable if its impulse response $h(t)$ satisfies one of the following conditions.

- Time-domain viewpoint:

$$\int_{-\infty}^{\infty} |h(t)| dt < \infty \quad (h(t) \text{ is absolutely integrable})$$

- Fourier transform viewpoint:

$$\left| \int_{-\infty}^{\infty} h(t) e^{-j\Omega t} dt \right| < \infty, \quad \forall \Omega \quad (\text{FT exists})$$

- Laplace transform viewpoint:

For causal rational $h(t)$, all poles of $H(s)$ are located in open left-half plane

DT-LTI systems:

- So far, we have reviewed continuous-time LTI systems:
 1. Eigenfunctions and FS (covers periodic signals)
 2. CTFT (extension to non-periodic signals)
 3. LT (“deluxe” version of the CTFT)
- Now, we discuss some counterparts for discrete-time LTI systems:
 - a. DTFT (analogous to CTFT)
 - b. Z-transform (analogous to LT)

We postpone our discussion of the DFT, the discrete-time counterpart to FS, until a bit later.

The DTFT:

- The *discrete-time Fourier transform* (DTFT) is defined as

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} \text{ for } \omega \in \mathbb{R}$$

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega})e^{j\omega n} d\omega,$$

where the notation “ $X(e^{j\omega})$ ” is consistent with the Z-transform. (More on this later...)

The lack of the subscript “ c ” distinguishes the DTFT and Z-transform “ $X(\cdot)$ ” from the CTFT and LT “ $X_c(\cdot)$ ”.

- Essentially, the DTFT gives a “continuous-frequency” characterization of DT signals in the same way that the CTFT gave a “continuous-frequency” characterization of CT signals. (The DTFT/CTFT relationship will be explained further in the context of *sampling*.)
- Important: the DTFT is 2π -periodic in ω :

$$X(e^{j\omega}) = X(e^{j(\omega+2\pi k)}) \text{ for any integer } k,$$

because the function $e^{j\omega}$ is itself 2π -periodic in ω .

In contrast, the CTFT $X_c(j\Omega)$ is not periodic in Ω !

The DTFT — important transform pairs:

- The DTFT is often written in “transform pair” notation

$$x[n] \xleftrightarrow{\text{DTFT}} X(e^{j\omega}).$$

- Some important examples of DTFT transform pairs are:

- Complex exponential:

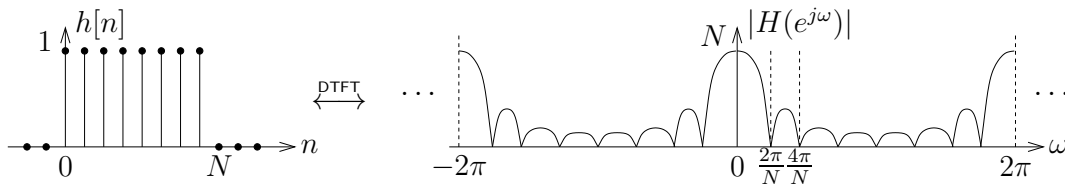
$$e^{j\omega_0 n} \xleftrightarrow{\text{DTFT}} 2\pi \sum_{k=-\infty}^{\infty} \delta(\omega - \omega_0 + 2\pi k)$$

- Kronecker delta:

$$\delta[n - d] \xleftrightarrow{\text{DTFT}} e^{-j\omega d}, d \in \mathbb{Z}$$

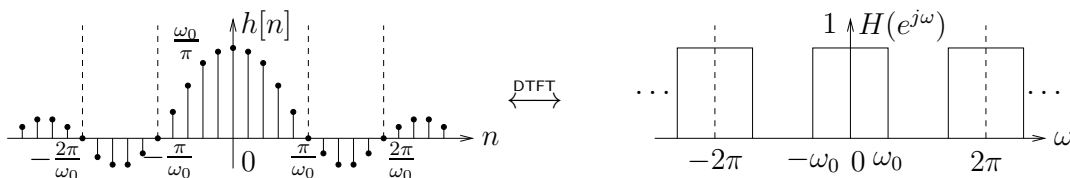
- Rectangle:

$$\begin{cases} 1 & n = 0, \dots, N-1 \\ 0 & \text{else} \end{cases} \xleftrightarrow{\text{DTFT}} e^{-j\omega \frac{N-1}{2}} \frac{\sin(\omega N/2)}{\sin(\omega/2)}$$



- Sinc (with $\omega_0 < \pi$):

$$\frac{\sin(\omega_0 n)}{\pi n} \xleftrightarrow{\text{DTFT}} \begin{cases} 1 & \text{if } |\omega + 2\pi k| < \omega_0, \text{ for } k \in \mathbb{Z} \\ 0 & \text{else} \end{cases}$$



The DTFT — important properties:

Below, assume $x[n] \xleftrightarrow{\text{DTFT}} X(e^{j\omega})$ and $w[n] \xleftrightarrow{\text{DTFT}} W(e^{j\omega})$.

1. Linearity:

$$\alpha x[n] + \beta w[n] \xleftrightarrow{\text{DTFT}} \alpha X(e^{j\omega}) + \beta W(e^{j\omega})$$

2. Conjugate symmetry:

$$x^*[n] = x[-n] \Leftrightarrow X(e^{j\omega}) \in \mathbb{R}$$

$$x[n] \in \mathbb{R} \Leftrightarrow X^*(e^{j\omega}) = X(e^{-j\omega})$$

3. Shift/modulation:

$$x[n-d] \xleftrightarrow{\text{DTFT}} e^{-j\omega d} X(e^{j\omega}), d \in \mathbb{Z}$$

$$e^{j\omega_0 n} x[n] \xleftrightarrow{\text{DTFT}} X(e^{j(\omega-\omega_0)})$$

4. Reversal:

$$x[-n] \xleftrightarrow{\text{DTFT}} X(e^{-j\omega})$$

5. Multiplication/convolution:

$$x[n]w[n] \xleftrightarrow{\text{DTFT}} \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\theta}) W(e^{j(\omega-\theta)}) d\theta$$

$$\sum_{m=-\infty}^{\infty} x[m]w[n-m] \xleftrightarrow{\text{DTFT}} X(e^{j\omega}) W(e^{j\omega})$$

6. Energy conservation (“Parseval’s theorem”):

$$\sum_{n=-\infty}^{\infty} |x[n]|^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} |X(e^{j\omega})|^2 d\omega$$

The DTFT — frequency domain convolution:

- The DTFT-domain convolution

$$X(e^{j\omega}) \circledast W(e^{j\omega}) = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\theta}) W(e^{j(\omega-\theta)}) d\theta$$

differs from “standard” convolution

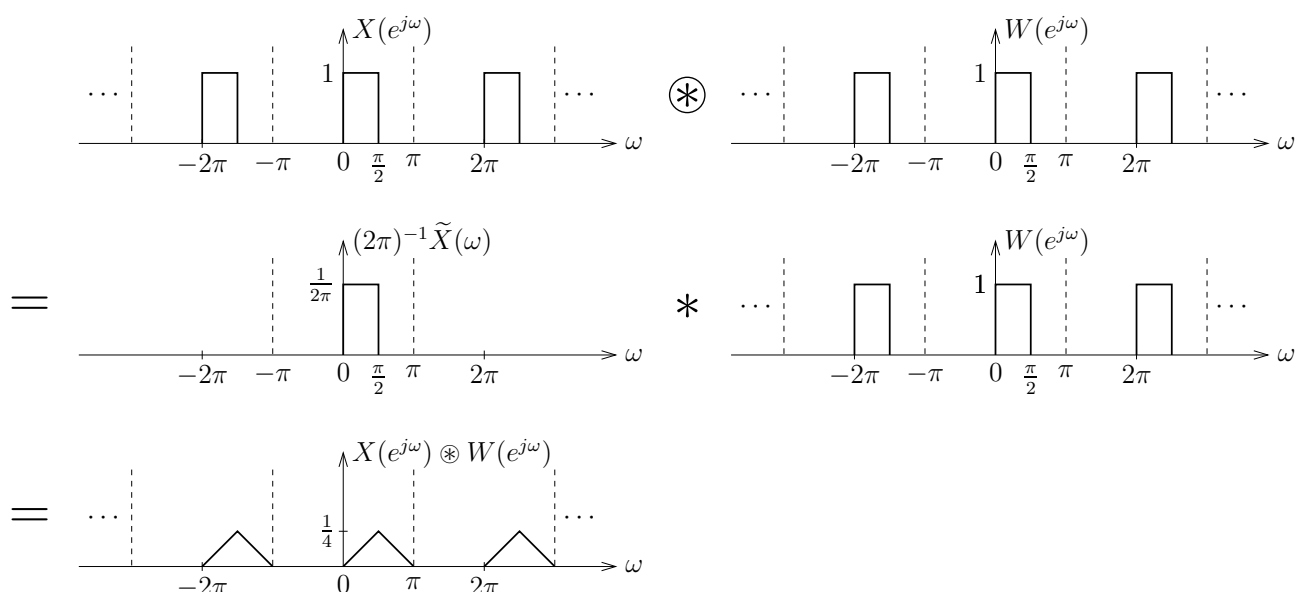
$$x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau) h(t - \tau) d\tau$$

in $\frac{1}{2\pi}$ -scaling but also in the limits of integration.

- But, if we define $\tilde{X}(\omega)$ as $X(e^{j\omega})$ with all content outside of $\omega \in (-\pi, \pi]$ suppressed, then we can write

$$\begin{aligned} X(e^{j\omega}) \circledast W(e^{j\omega}) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \tilde{X}(\theta) W(e^{j(\omega-\theta)}) d\theta \\ &= \frac{1}{2\pi} \tilde{X}(\omega) * W(e^{j\omega}) \end{aligned}$$

- Example:



Example — DTFT symmetry property:

Example — DTFT convolution property:

$$\begin{aligned}\mathcal{F}_{\text{DTFT}}\left\{\sum_m x[m]w[n-m]\right\} \\&= \sum_n \left(\sum_m x[m]w[n-m]\right)e^{-j\omega n} \\&= \sum_m x[m] \sum_n w[n-m]e^{-j\omega n} \\&= \sum_m x[m] \sum_q w[q]e^{-j\omega(q+m)}\end{aligned}$$

since $q = n - m$ and summation limits don't change

$$\begin{aligned}&= \sum_m x[m]e^{-j\omega m} \sum_q w[q]e^{-j\omega q} \\&= X(e^{j\omega}) W(e^{j\omega})\end{aligned}$$

Example — DTFT multiplication property:

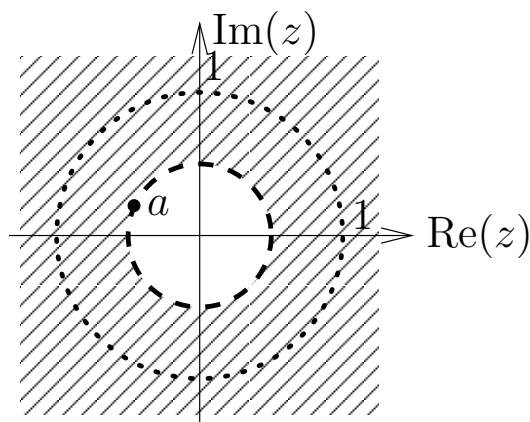
$$\begin{aligned} & \mathcal{F}_{\text{DTFT}}^{-1} \left\{ \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\xi}) W(e^{j(\omega-\xi)}) d\xi \right\} \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} \left(\frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\xi}) W(e^{j(\omega-\xi)}) d\xi \right) e^{j\omega n} d\omega \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\xi}) \left(\frac{1}{2\pi} \int_{-\pi}^{\pi} W(e^{j(\omega-\xi)}) e^{j\omega n} d\omega \right) d\xi \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\xi}) \left(\frac{1}{2\pi} \int_{-\pi-\xi}^{\pi-\xi} W(e^{j\theta}) e^{j(\theta+\xi)n} d\theta \right) d\xi \\ & \qquad \qquad \qquad \text{via } \theta \triangleq \omega - \xi \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\xi}) e^{j\xi n} \left(\frac{1}{2\pi} \int_{-\pi}^{\pi} W(e^{j\theta}) e^{j\theta n} d\theta \right) d\xi \\ & \qquad \qquad \qquad \text{since DTFT is } 2\pi\text{-periodic} \\ &= x[n]w[n] \\ & \qquad \qquad \qquad \text{via IDTFT definition} \end{aligned}$$

The Z-transform:

- Recall the 2-sided *Z-transform* (ZT) of $x[n]$:

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n}, \text{ for } z \in \mathbb{C} \text{ such that } |X(z)| < \infty.$$

- The DTFT can be considered as a special case of the ZT for which $z = e^{j\omega}$ with $\omega \in \mathbb{R}$, and that is why we write the DTFT of $x[n]$ as “ $X(e^{j\omega})$.”
- The ZT of a given $x[n]$ is defined only for those $z \in \mathbb{C}$ such that $|X(z)| < \infty$. This portion of the complex plane is known as the “*region of convergence* (ROC) of $X(z)$.”
For example, if $x[n] = a^n u[n]$, where $a \in \mathbb{C}$ and $u[n]$ is the step sequence ($u[n] = 1$ for $n \geq 0$ and $u[n] = 0$ otherwise), then one can show that the ROC is $\{z \in \mathbb{C} : |z| > |a|\}$.



- Like the DTFT, the utility of the ZT stems from the fact that the convolution infinite-sum

$$y[n] = \sum_{m=-\infty}^{\infty} h[m]x[n-m]$$

simplifies to the algebraic expression

$$Y(z) = H(z)X(z)$$

.

- For many signals $x[n]$ (e.g., impulse, step, ramp, geometric signals $a^n u[n]$), there exist convenient closed-form expressions for $X(z)$. Moreover, when $Y(z)$ is rational, its inverse ZT can be calculated using *partial fraction expansion* (PFE) and table-lookup.
- The *advantage* of the ZT vs. the DTFT is that there is a much larger class of signals with elegant transforms. The *disadvantage* of the ZT vs. the DTFT is that the inverse transform step (requiring PFE or contour integration) is more complicated.

In this course, we focus on the DTFT but use the ZT when building block diagrams (since “ z^{-1} ” describes a system that delays its input by one sample).

$$y[n] = ax[n-1] + bx[n-2]$$

Example 1 — LTI transfer functions:

The DTFT leads to a convenient algebra for LTI systems:

- For an LTI system, the impulse response $h[n]$ characterizes the I/O behavior:

$$x[n] \rightarrow \boxed{\mathcal{H}} \rightarrow x[n] * h[n]$$

- The DTFT convolution property says

$$x[n] * h[n] \xleftrightarrow{\text{DTFT}} X(e^{j\omega})H(e^{j\omega}),$$

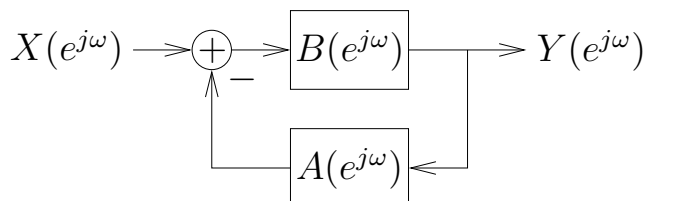
implying that LTI I/O behavior can also be expressed as

$$X(e^{j\omega}) \rightarrow \boxed{\mathcal{H}} \rightarrow X(e^{j\omega})H(e^{j\omega})$$

i.e., *multiplication* by “transfer function” $H(e^{j\omega})$:

$$H(e^{j\omega}) \xleftrightarrow{\text{DTFT}} h[n]$$

- In this way, the DTFT facilitates *algebraic* analysis of networks of interconnected systems!


$$\frac{Y(e^{j\omega})}{X(e^{j\omega})} = \frac{B(e^{j\omega})}{1 + A(e^{j\omega})B(e^{j\omega})}$$

Example 2 — ARMA transfer function:

Say that we have an “autoregressive moving average” system:

$$y[n] = \sum_{m=0}^M b_m x[n-m] - \sum_{k=1}^N a_k y[n-k].$$

It's not difficult to show that such systems are LTI.

We now derive the *ARMA transfer function*:

$$\begin{aligned} Y(e^{j\omega}) &\triangleq \mathcal{F}_{\text{DTFT}}\{y[n]\} \\ &= \sum_{m=0}^M b_m \mathcal{F}_{\text{DTFT}}\{x[n-m]\} - \sum_{k=1}^N a_k \mathcal{F}_{\text{DTFT}}\{y[n-k]\} \quad (\text{linearity}) \\ &= \sum_{m=0}^M b_m e^{-j\omega m} X(e^{j\omega}) - \sum_{k=1}^N a_k e^{-j\omega k} Y(e^{j\omega}) \quad (\text{delay}) \end{aligned}$$

so that

$$Y(e^{j\omega}) \left(1 + \sum_{k=1}^N a_k e^{-j\omega k} \right) = X(e^{j\omega}) \sum_{m=0}^M b_m e^{-j\omega m}$$

and thus

$$H(e^{j\omega}) = \frac{Y(e^{j\omega})}{X(e^{j\omega})} = \frac{\sum_{m=0}^M b_m e^{-j\omega m}}{1 + \sum_{k=1}^N a_k e^{-j\omega k}}.$$

Example 3 — Direct form II realization:

Example 4 — LTI response to real sinusoid:

What is the response of an LTI system to a real sinusoid?

$$x[n] = \cos(\omega_0 n + \phi) \rightarrow \boxed{\mathcal{H}} \rightarrow y[n]$$

LTI implies that I/O behavior is characterized by transfer fxn:

$$Y(e^{j\omega}) = H(e^{j\omega})X(e^{j\omega})$$

Using Euler's expansion $\cos(\theta) = \frac{1}{2}(e^{j\theta} + e^{-j\theta})$,

$$X(e^{j\omega}) = \mathcal{F}_{\text{DTFT}} \left\{ \frac{1}{2} (e^{j\omega_0 n + j\phi} + e^{-j\omega_0 n - j\phi}) \right\}$$

Using $e^{\mp j\omega_0 n} \xleftrightarrow{\text{DTFT}} 2\pi \sum_k \delta(\omega \pm \omega_0 + 2\pi k)$,

$$X(e^{j\omega}) = \frac{2\pi}{2} \sum_k \left[e^{j\phi} \delta(\omega - \omega_0 + 2\pi k) + e^{-j\phi} \delta(\omega + \omega_0 + 2\pi k) \right]$$

Thus output is

$$Y(e^{j\omega}) = \pi \sum_k \left(e^{j\phi} \underbrace{H(e^{j\omega})}_{H(e^{j\omega_0})} \delta(\omega - \omega_0 + 2\pi k) + e^{-j\phi} \underbrace{H(e^{j\omega})}_{H(e^{-j\omega_0})} \delta(\omega + \omega_0 + 2\pi k) \right)$$

Using same DTFT pair again,

$$y[n] = \frac{1}{2} \left(e^{j\phi} H(e^{j\omega_0}) e^{j\omega_0 n} + e^{-j\phi} H(e^{-j\omega_0}) e^{-j\omega_0 n} \right)$$

If $h[n] \in \mathbb{R}$, then $H(e^{-j\omega_0}) = H^*(e^{j\omega_0})$. In this case, writing $H(e^{j\omega_0}) = |H(e^{j\omega_0})| e^{j\angle H(e^{j\omega_0})}$, we get

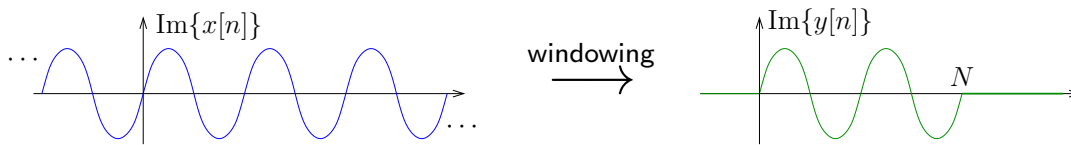
$$\begin{aligned} y[n] &= |H(e^{j\omega_0})| \frac{1}{2} \left(e^{j[\omega_0 n + \phi + \angle H(e^{j\omega_0})]} + e^{-j[\omega_0 n + \phi + \angle H(e^{j\omega_0})]} \right) \\ &= |H(e^{j\omega_0})| \cos(\omega_0 n + \phi + \angle H(e^{j\omega_0})) \end{aligned}$$

Real LTI sys changes only gain & phase of real sinusoid!

Example 5 — windowing a sinusoid:

Say $x[n] = e^{j\omega_0 n} \forall n \in \mathbb{Z}$ and $w[n] = \begin{cases} 1 & n = 0, \dots, N-1 \\ 0 & \text{else} \end{cases}$

and $y[n] = w[n]x[n]$:



What is $Y(e^{j\omega})$?

Recall that $X(e^{j\omega}) = 2\pi \sum_{k=-\infty}^{\infty} \delta(\omega - \omega_0 + 2\pi k)$.

The DTFT multiplication property then says

$$\begin{aligned}
 Y(e^{j\omega}) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\xi}) W(e^{j(\omega-\xi)}) d\xi \\
 &= \sum_{k=-\infty}^{\infty} \int_{-\pi}^{\pi} \delta(\xi - \omega_0 + 2\pi k) W(e^{j(\omega-\xi)}) d\xi \\
 &= \int_{-\pi}^{\pi} \delta(\xi - \omega_0) W(e^{j(\omega-\xi)}) d\xi \quad \text{assuming } \omega_0 \in (-\pi, \pi] \\
 &= W(e^{j(\omega-\omega_0)}).
 \end{aligned}$$

For the (causal N -length) rectangular “window” $w[n]$, know

$$W(e^{j\omega}) = e^{-j\omega \frac{N-1}{2}} \frac{\sin(\omega N/2)}{\sin(\omega/2)}.$$

